Extreme magnetic fields around black holes
@Astrolunch UCSB

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Event Horizon Telescope

BLACK HOLE INITIATIVE

HARVARD UNIVERSITY

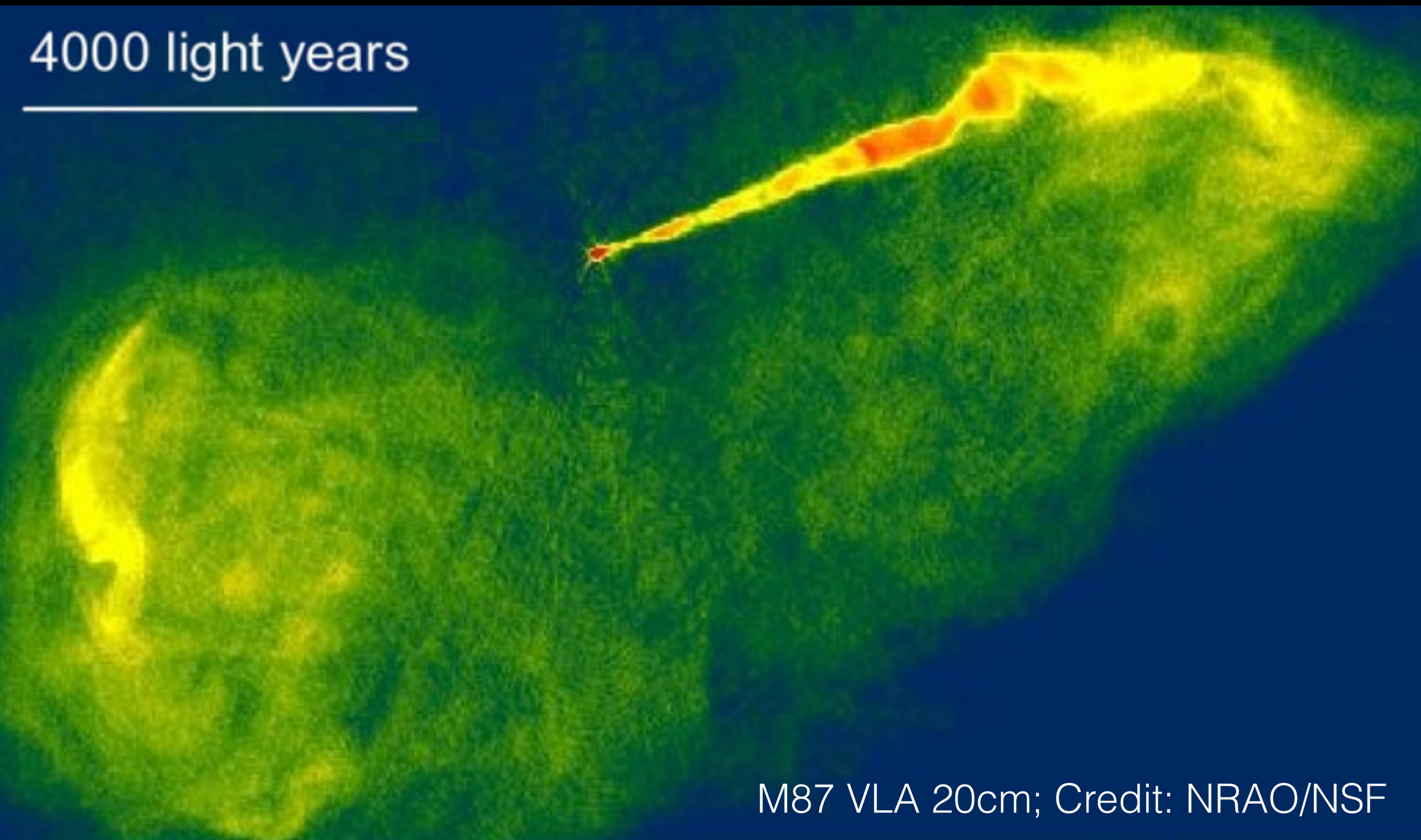
UNIVERSITY OF MARYLAND 18 56

NSF

compute canada | calcul canada

summit

Jets in nature: M87



M87 VLA 20cm; Credit: NRAO/NSF

Jets in nature: M87

$\dot{M}c^2$
Accretion

4000 light years

M87 VLA 20cm; Credit: NRAO/NSF

Jets in nature: M87

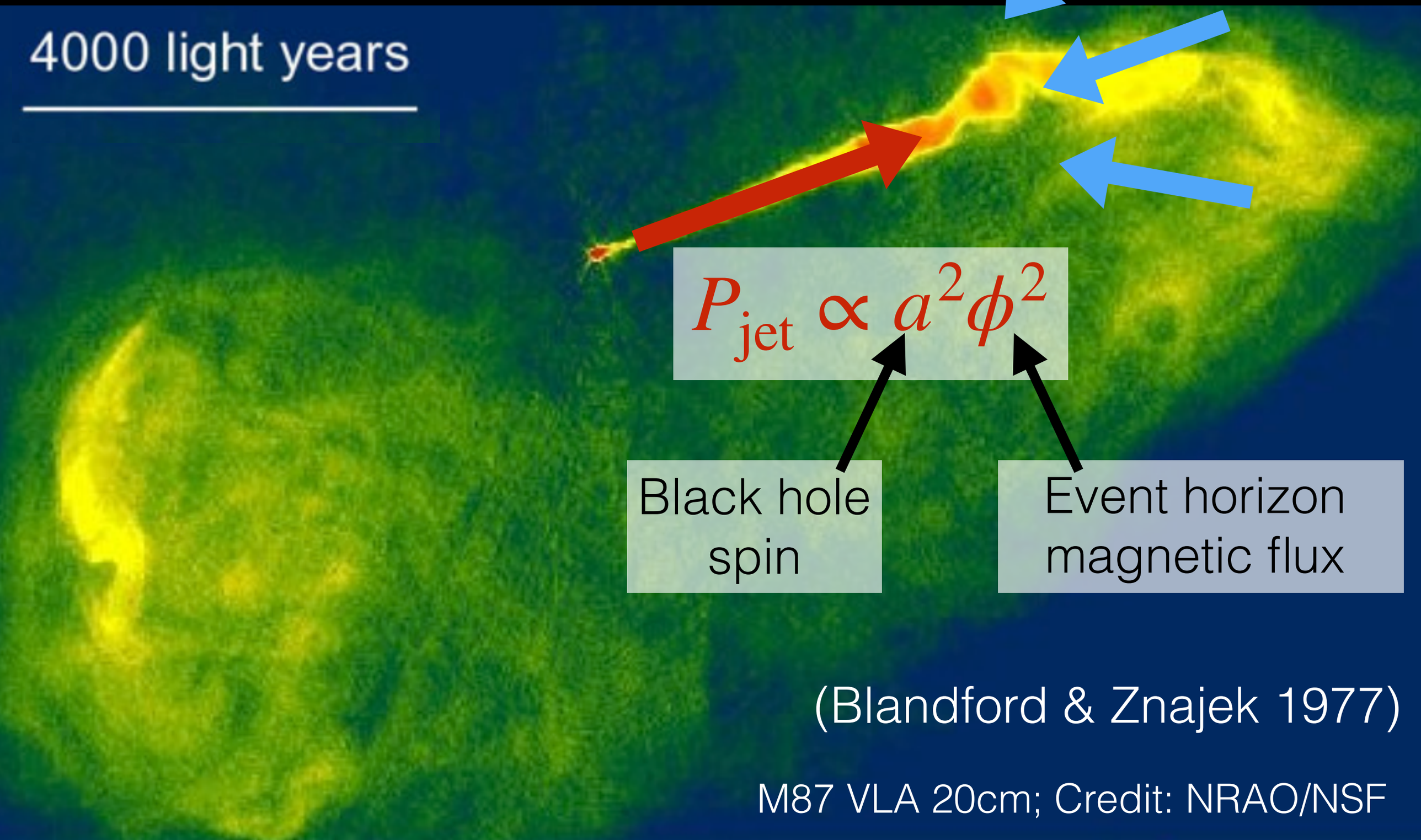
$\dot{M}c^2$
Accretion

4000 light years

$P_{\rm jet}$

M87 VLA 20cm; Credit: NRAO/NSF

Jets in nature: M87



(Blandford & Znajek 1977)

M87 VLA 20cm; Credit: NRAO/NSF

Jets in nature: M87

$\dot{M}c^2$
Accretion

4000 light years

$P_{\rm jet} \propto a^2 \phi^2$

Black hole spin

Event horizon magnetic flux



(Blandford & Znajek 1977)

M87 VLA 20cm; Credit: NRAO/NSF

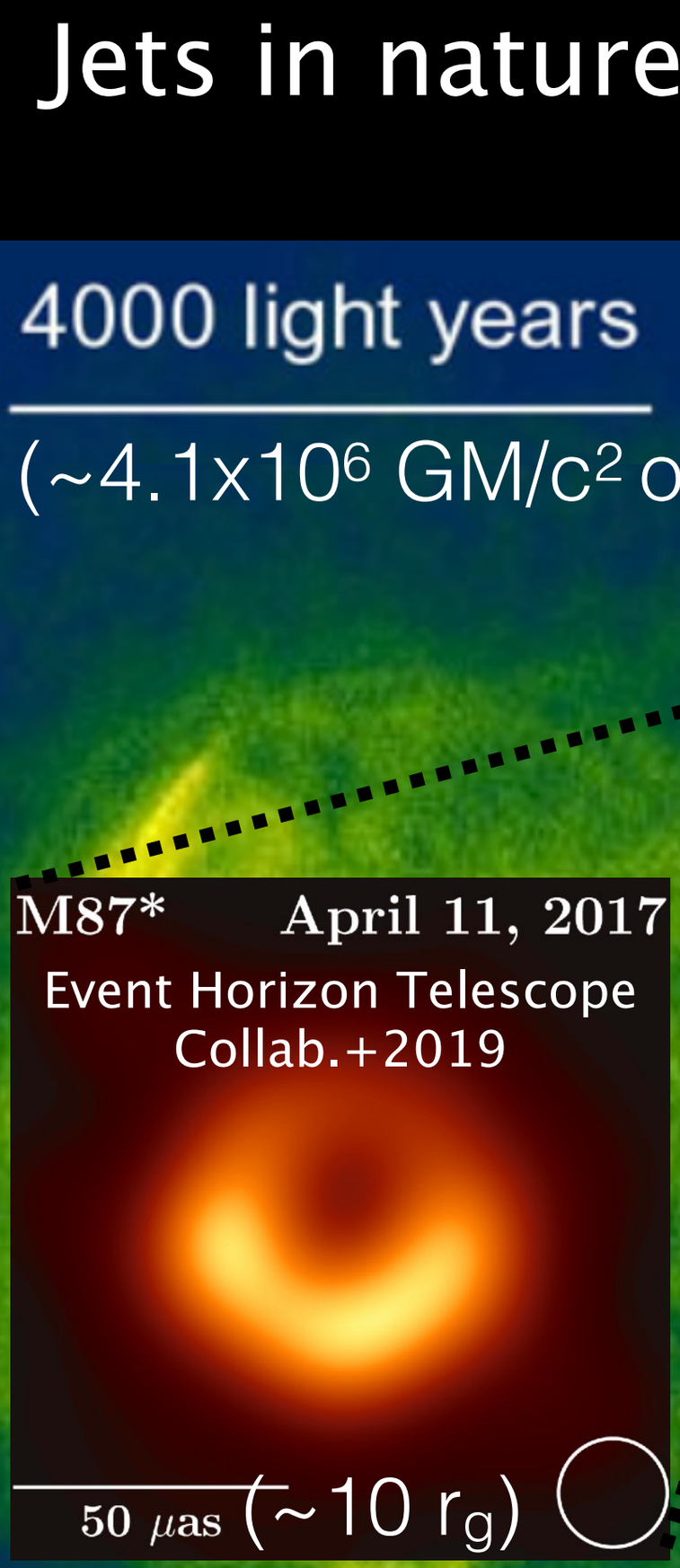
Jets in nature: M87

$\dot{M}c^2$
Accretion

4000 light years

(~4.1x10^6 GM/c^2 or r_g)

$P_{\rm jet} \propto a^2 \phi^2$



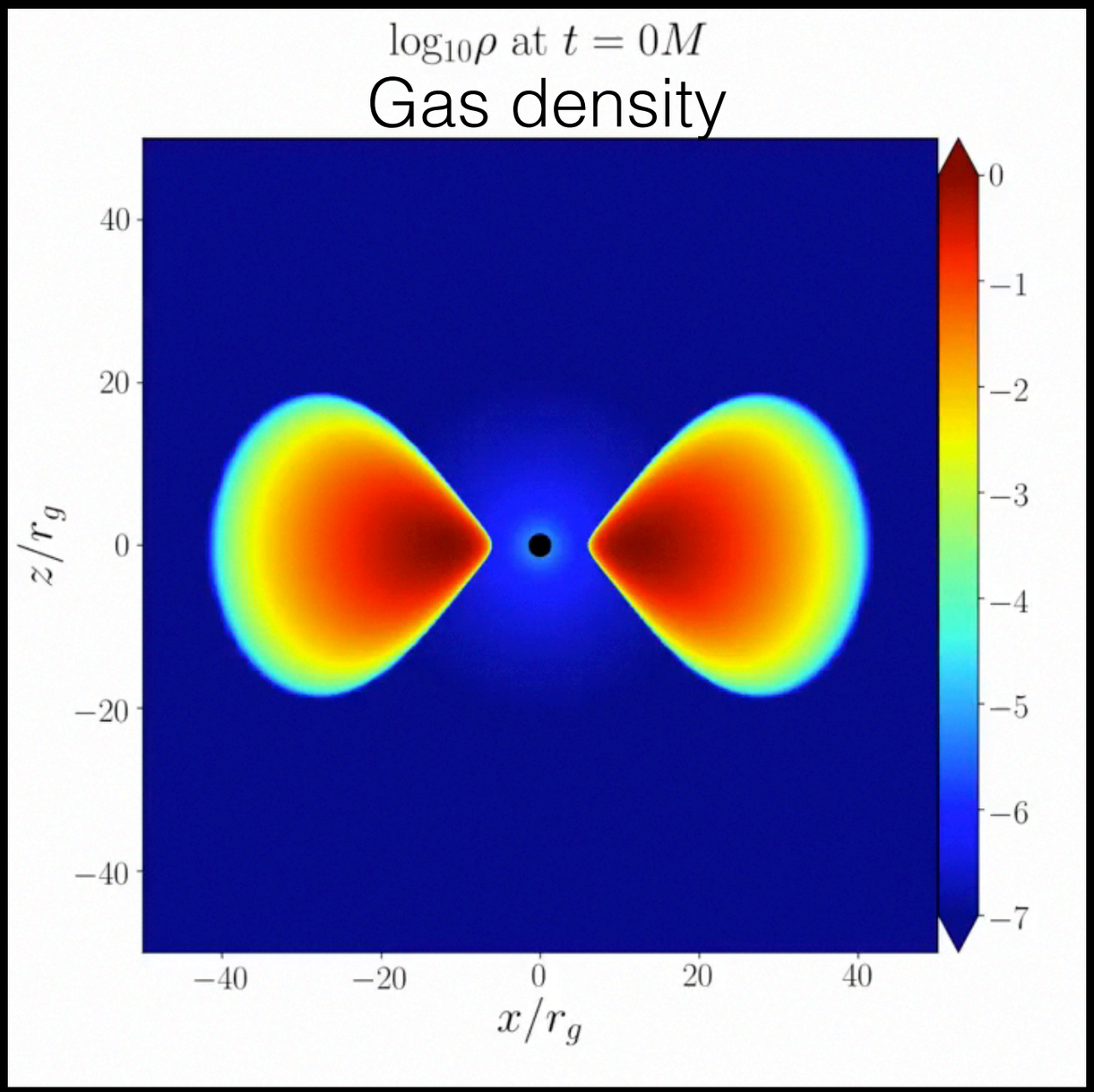
~7-8 orders of separation in length-scale!!

(Angles-Alcazar+, Cho+, Guo+, Hopkins+, KC+, Lalakos+, Ressler+, many)

M87 VLA 20cm; Credit: NRAO/NSF

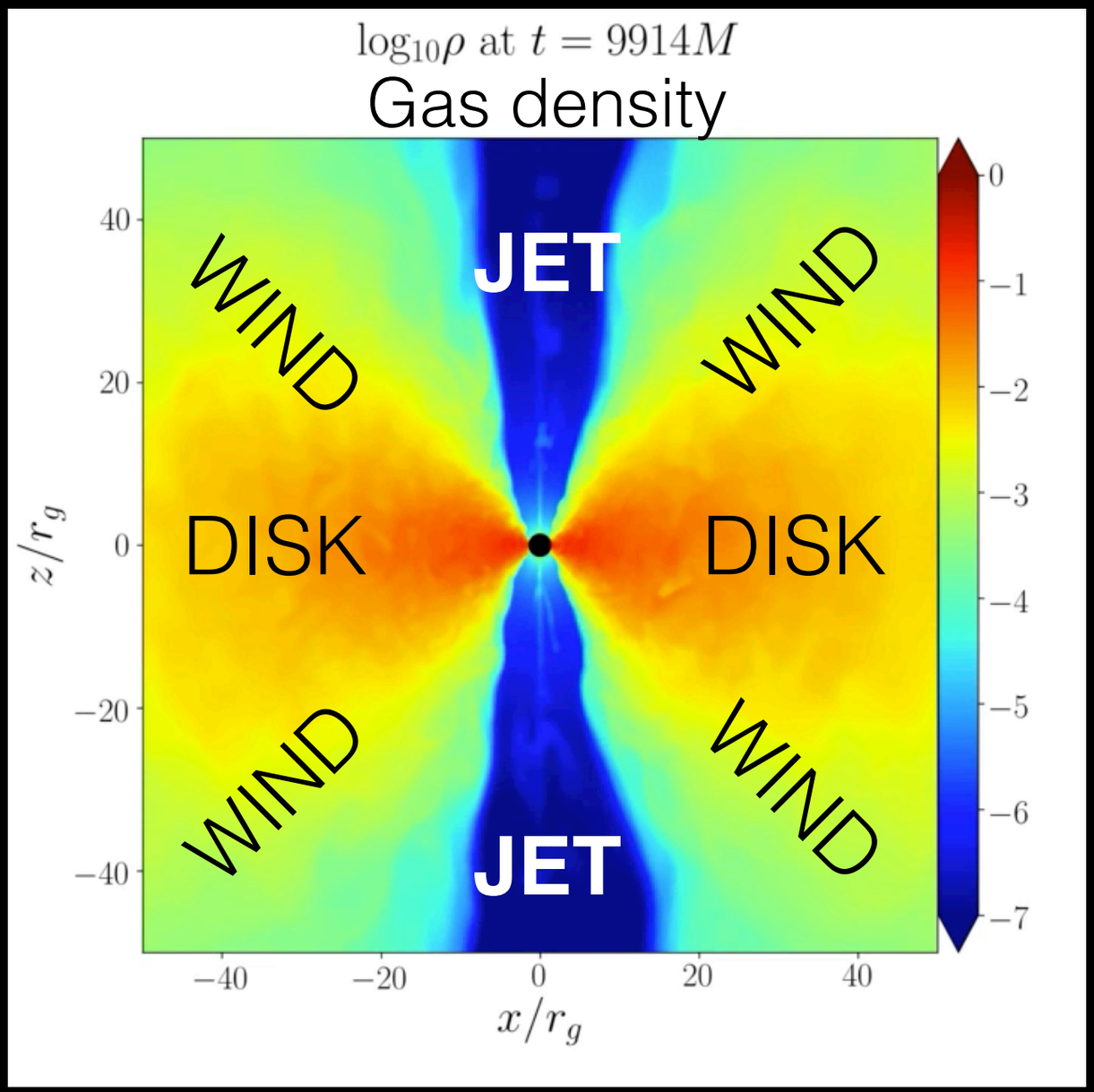
Jet propagation from first principles

Magnetized torus



Jet propagation from first principles

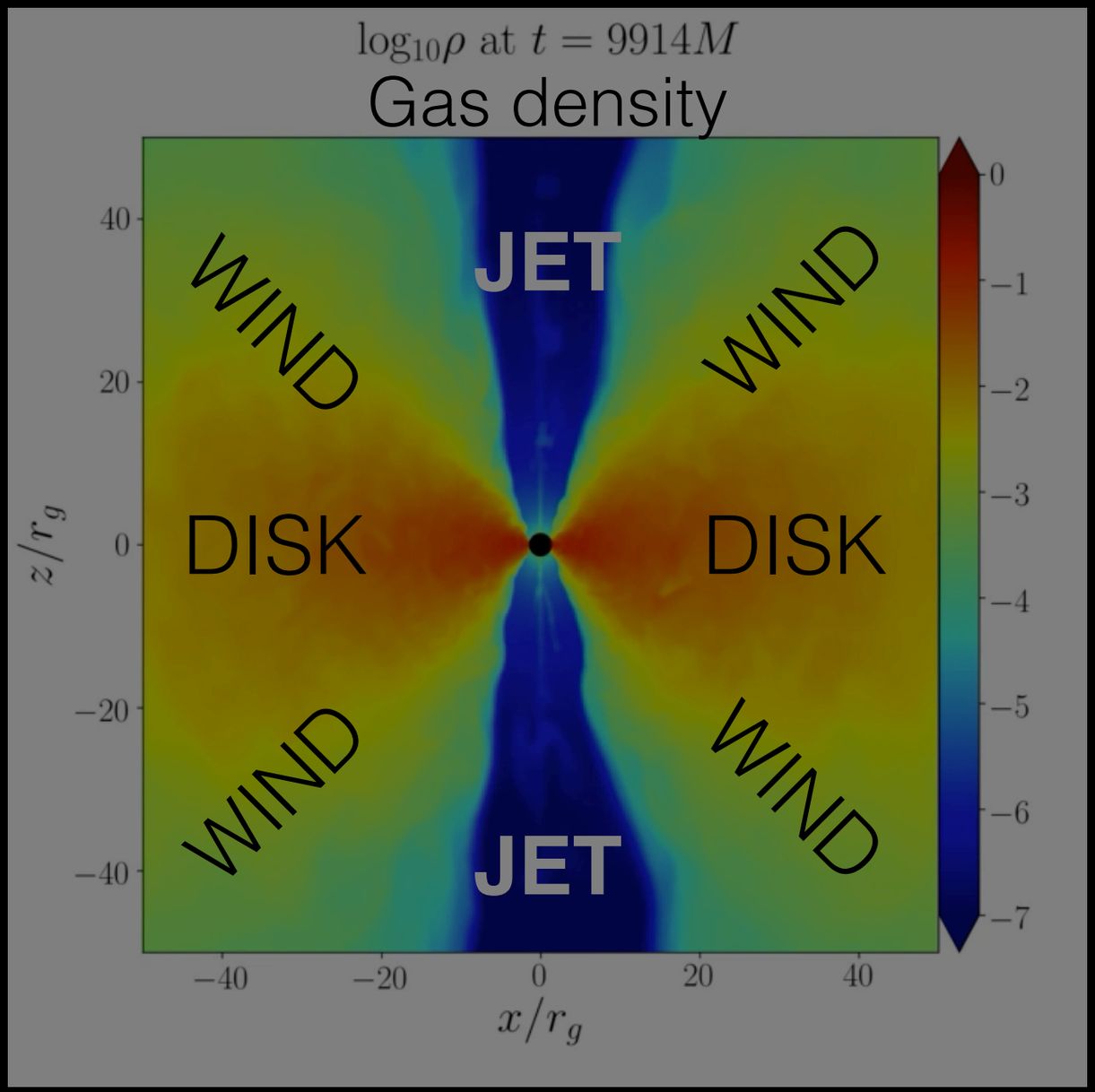
Magnetized torus



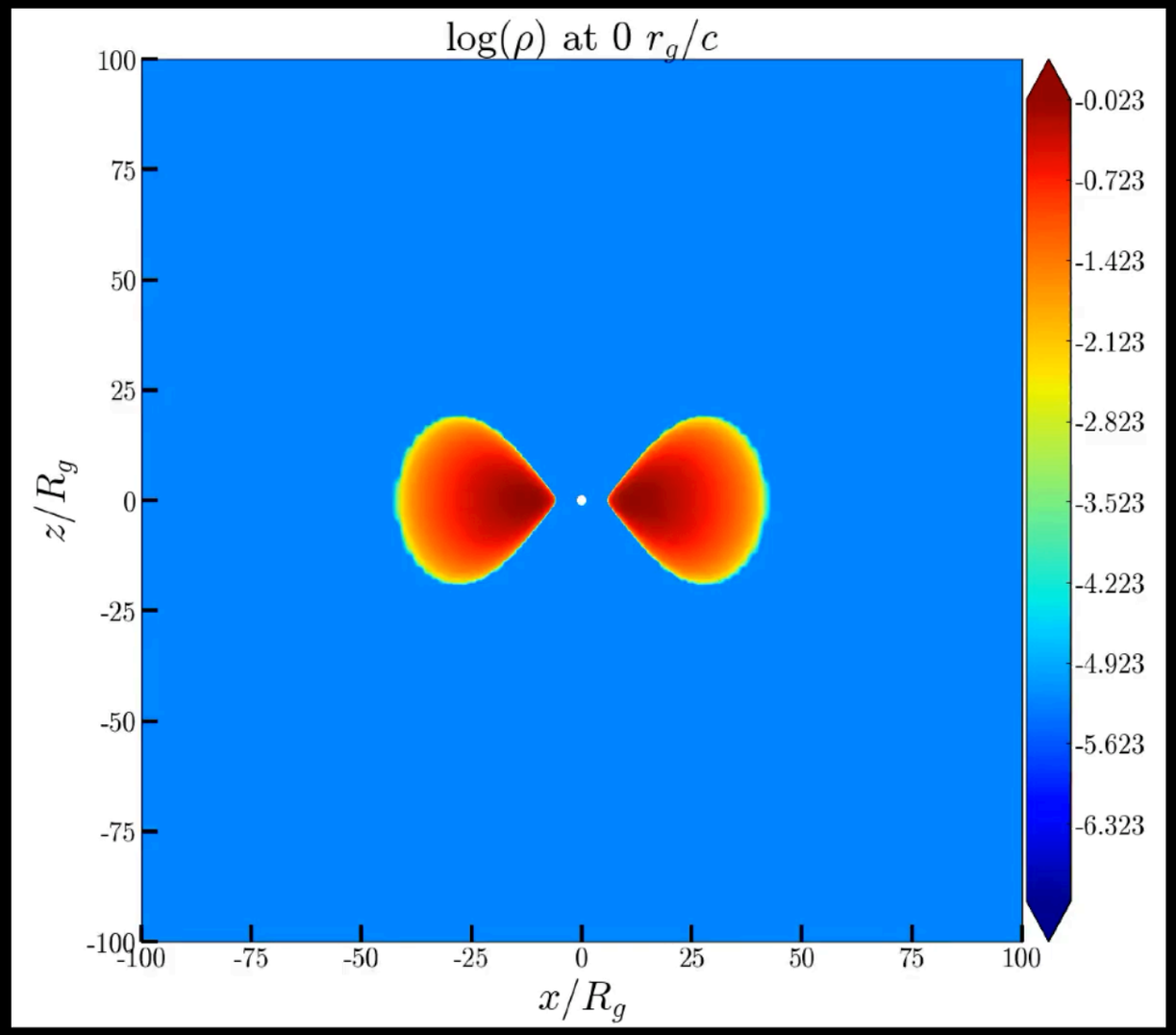
- Turbulent, magnetized disk
- Highly magnetized, fast, near-vacuum outflow (jet)
- Slow, gas-rich disk outflow (wind)

Jet propagation from first principles

Magnetized torus



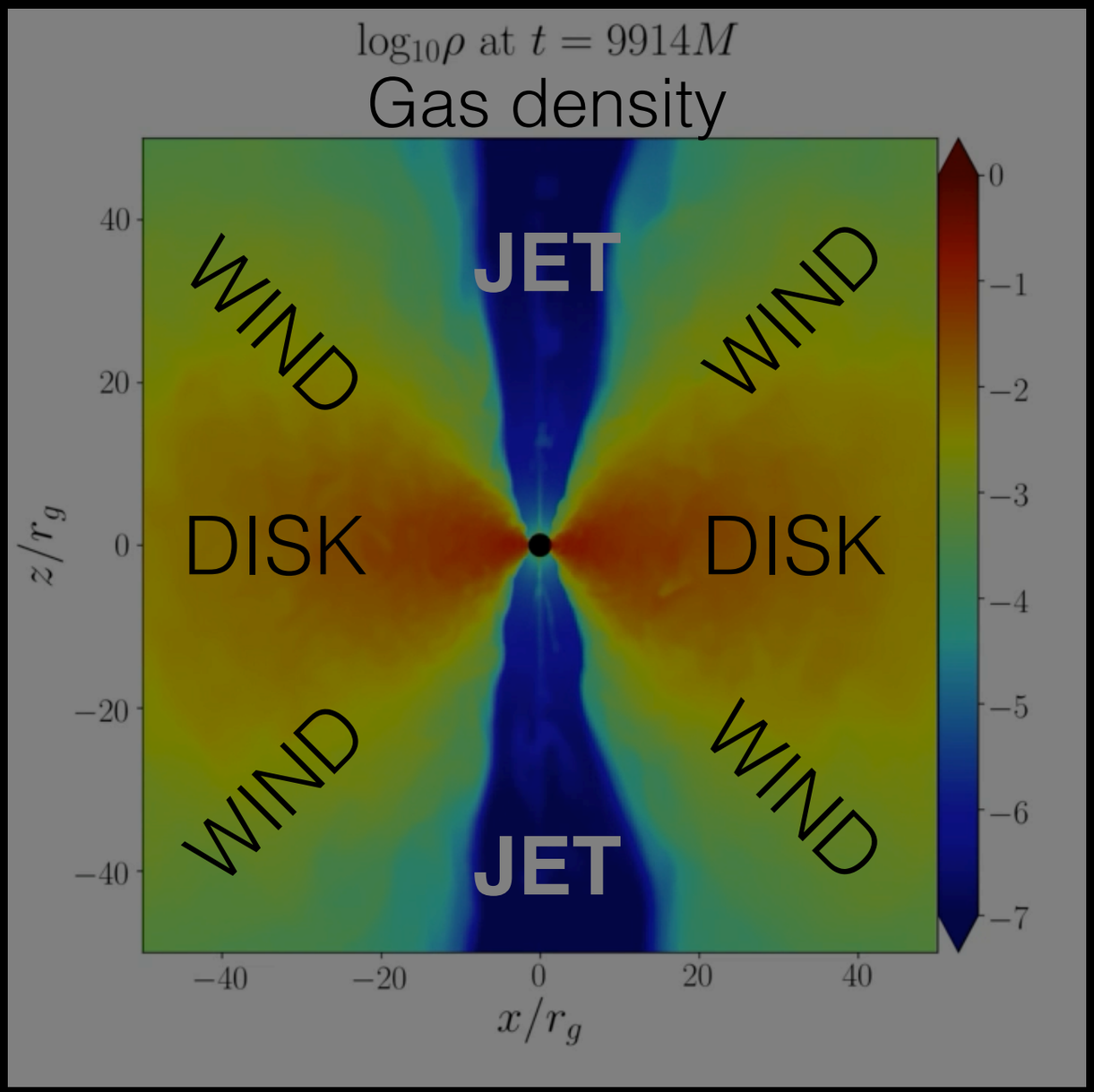
Magnetized torus in a medium



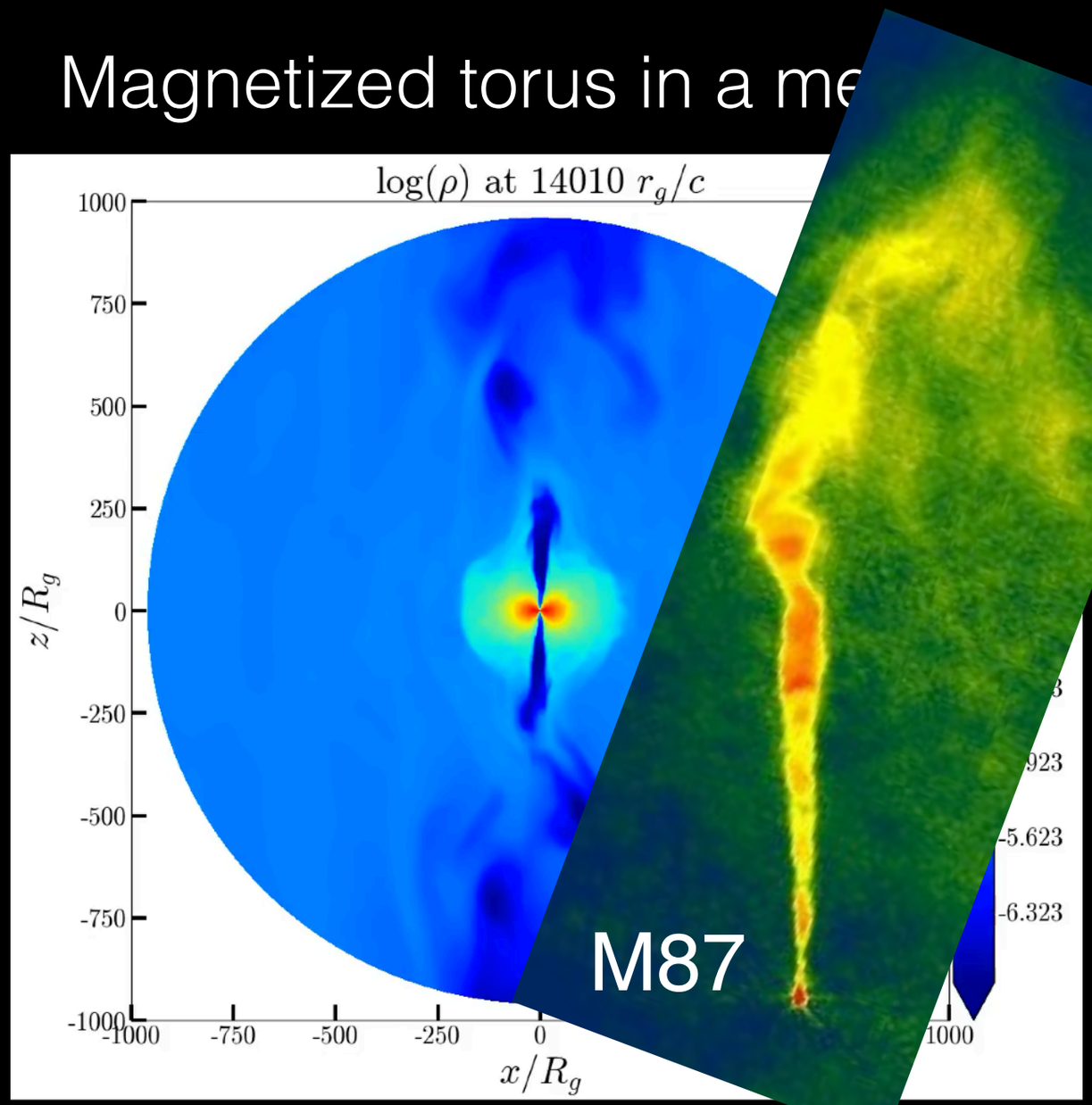
The jet can kink and fall apart

Jet propagation from first principles

Magnetized torus



Magnetized torus in a me

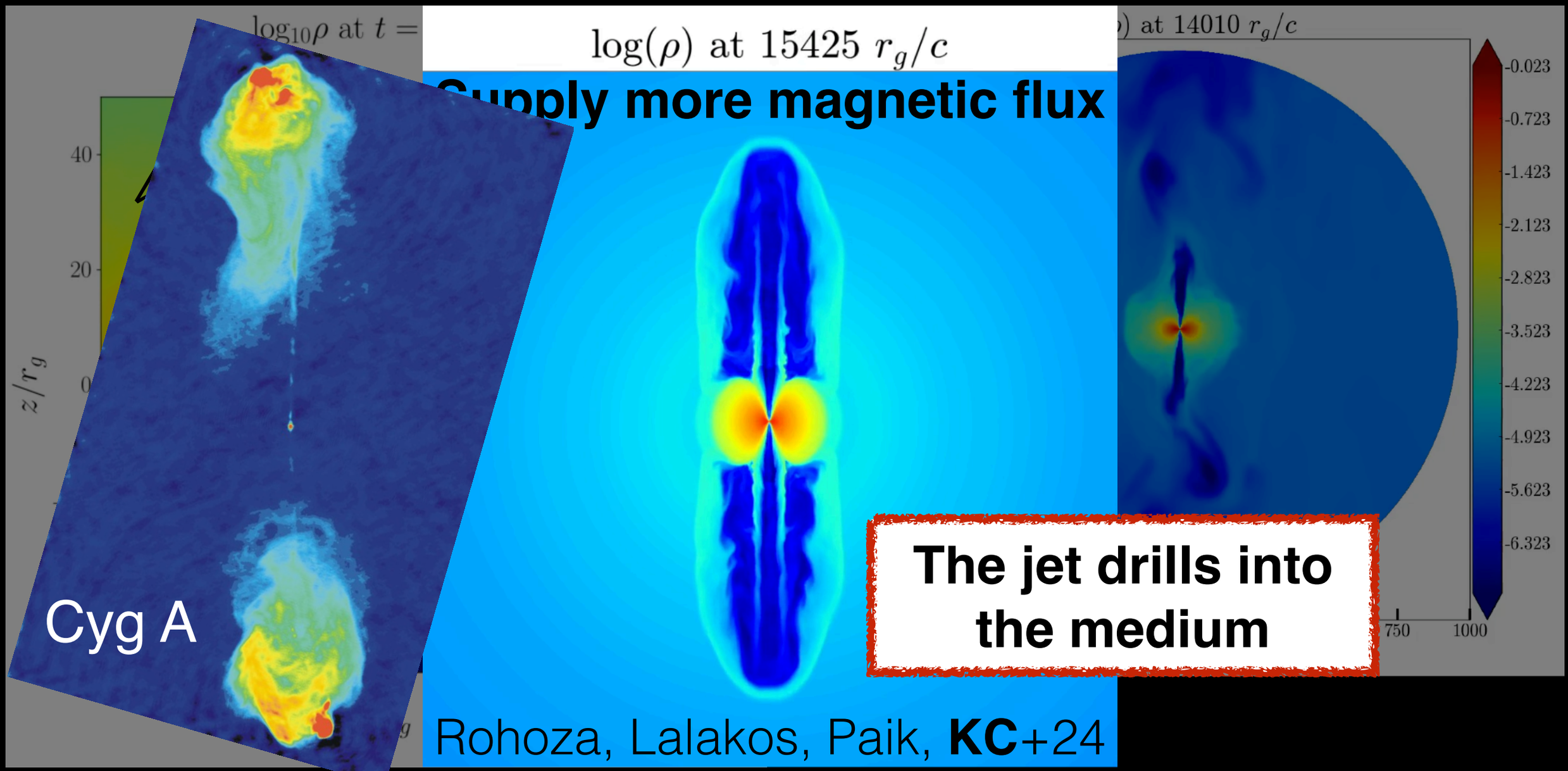


The jet can kink and fall apart

Strong magnetic fields launch stable jets

Magnetized torus

Magnetized torus in a medium



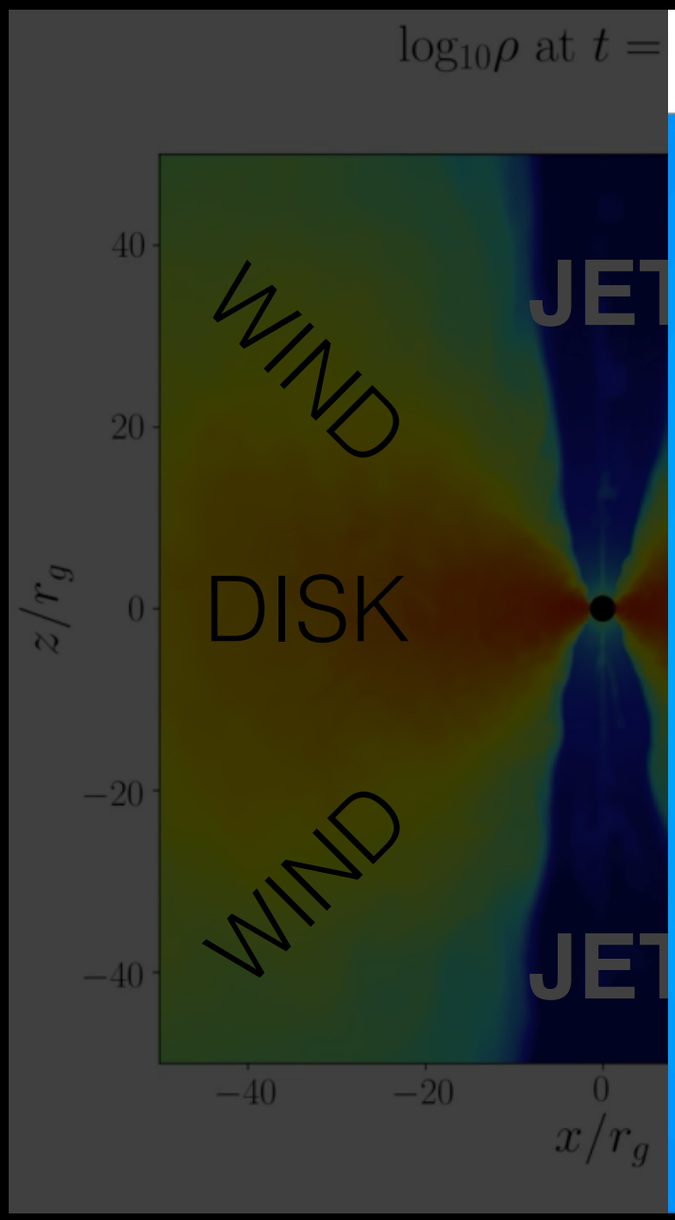
Rohoza, Lalakos, Paik, **KC**+24

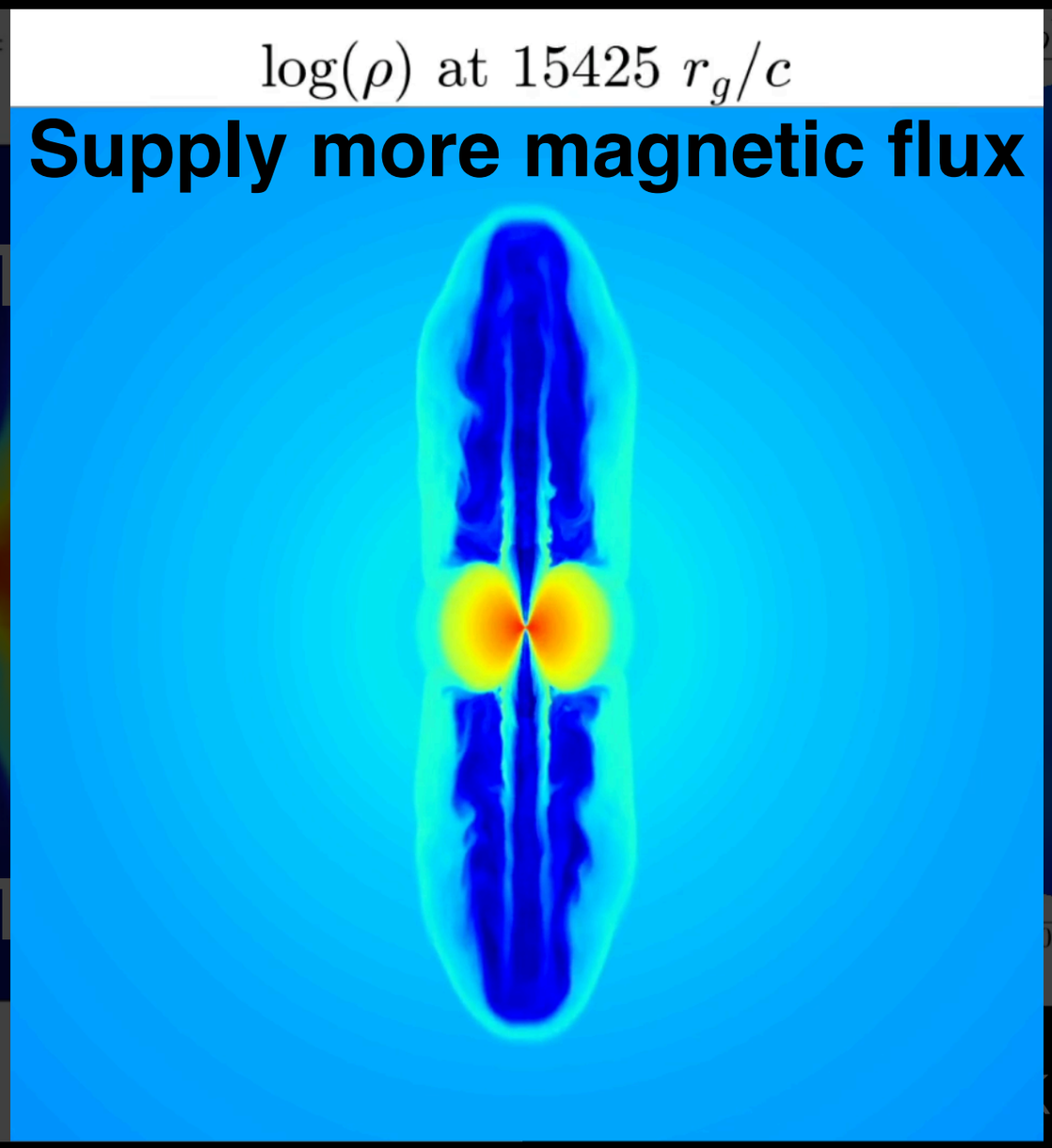
Lalakos,..,**KC**+23

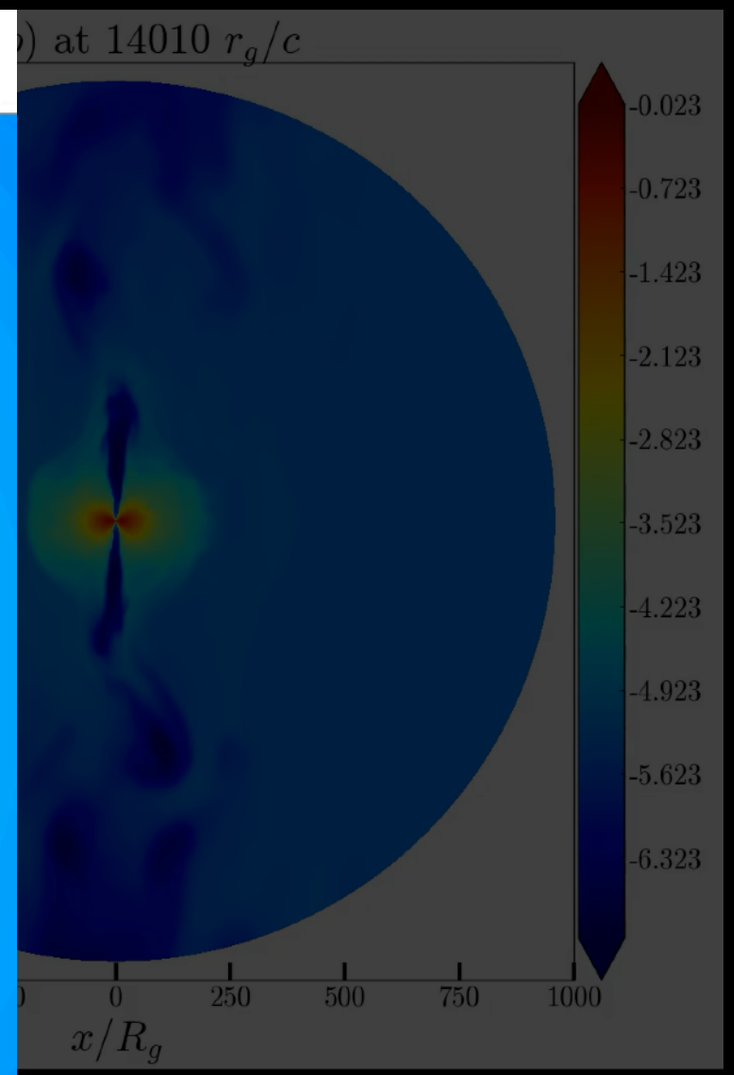
Strong magnetic fields launch stable jets

Magnetized torus

Magnetized torus in a medium





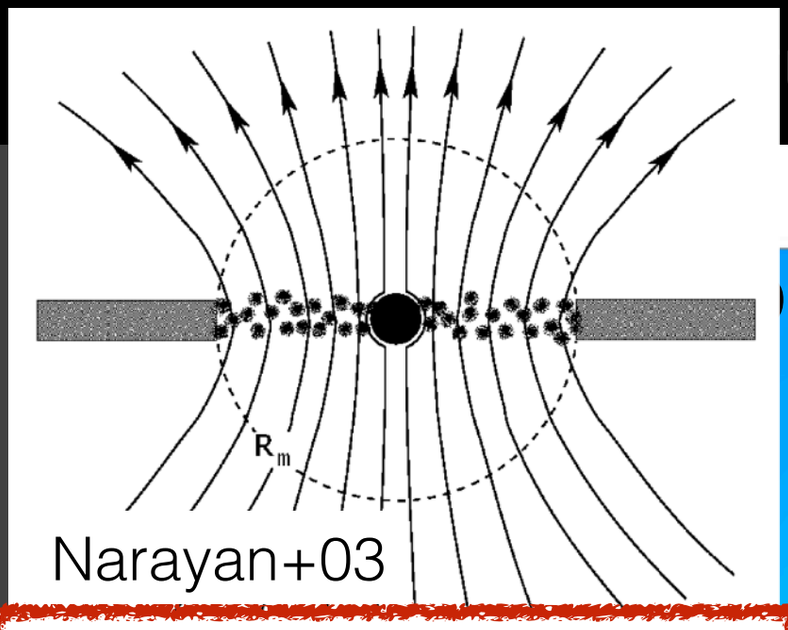


kink and fall apart

Magnetically Arrested Disks (or MADs)

us

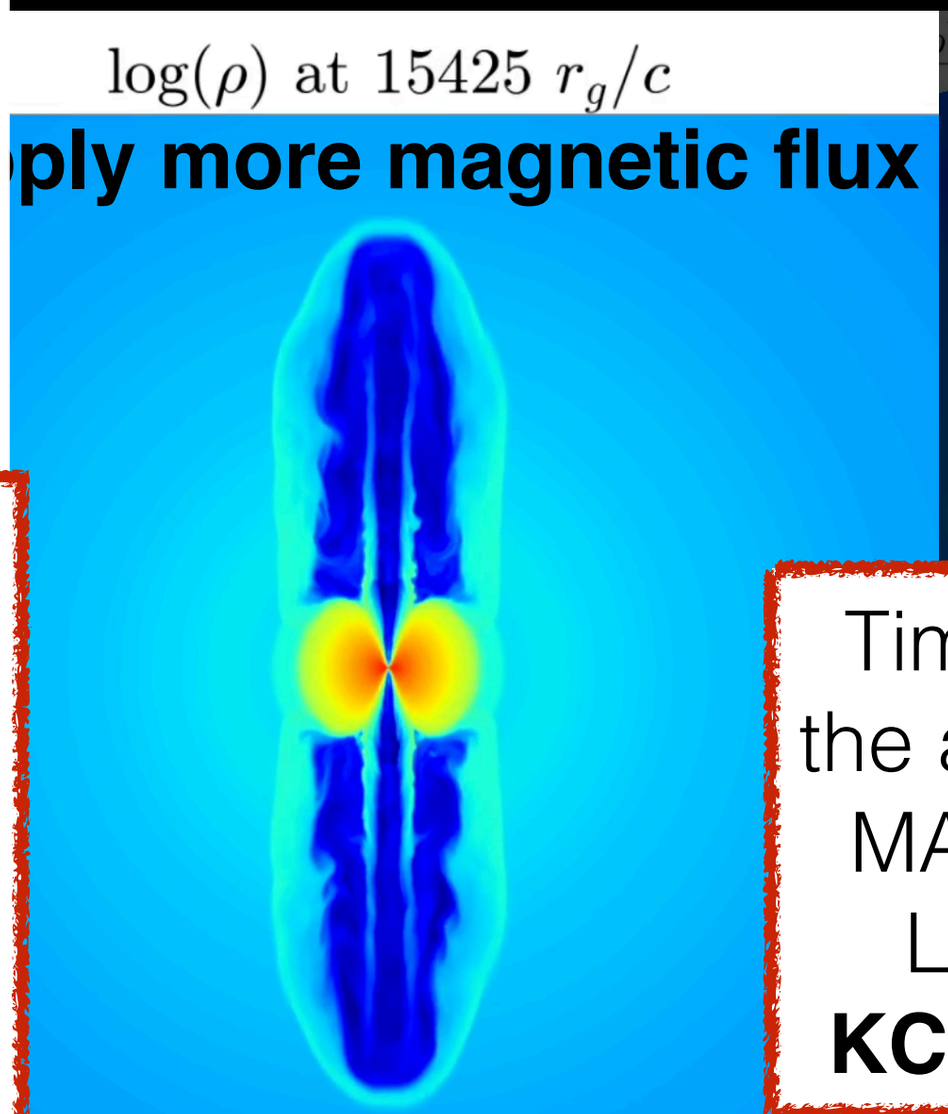
Magnetized torus in a medium



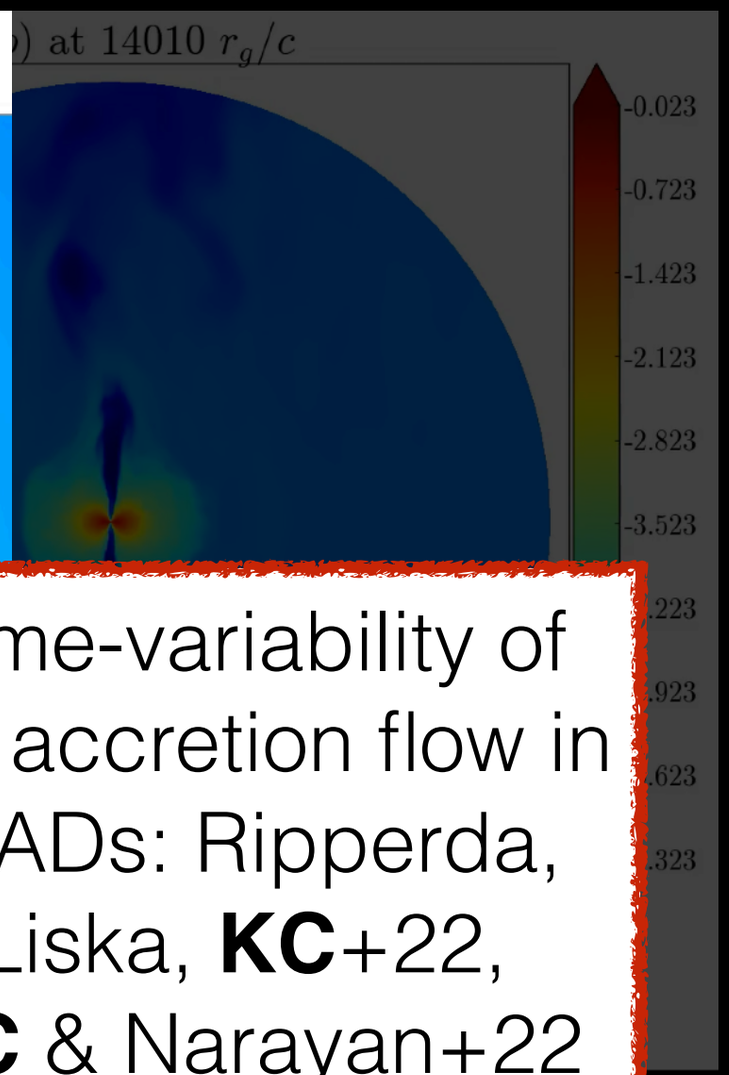
Narayan+03

Magnetic pressure can overcome gravity

(Bisnovatyi-Kogan+74, Igumenshchev+03, Narayan+03, Tchekhovskoy+11)



pply more magnetic flux

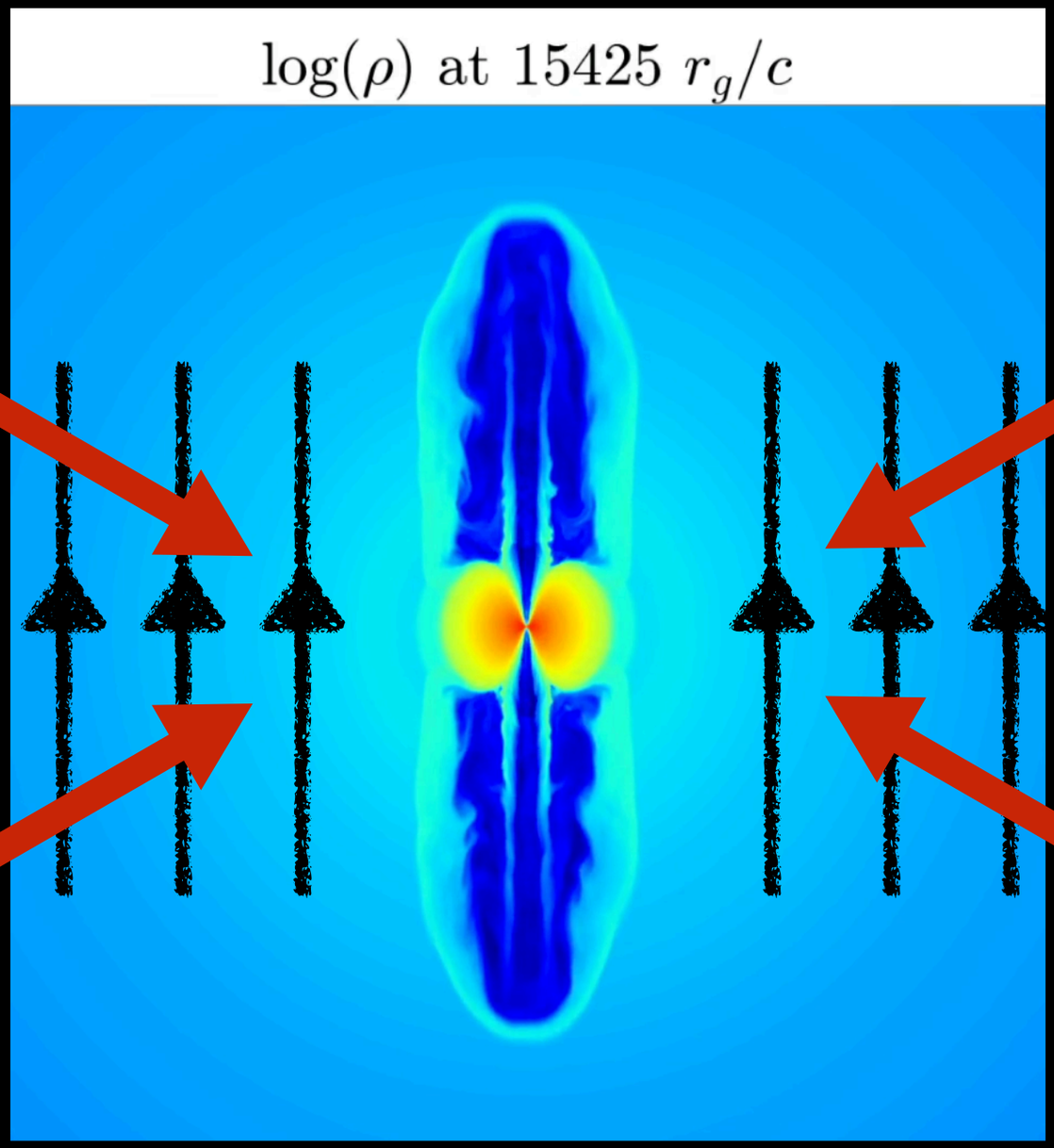


Time-variability of the accretion flow in MADs: Ripperda, Liska, **KC**+22, **KC** & Narayan+22

tink and fall apart

MADs: supply of magnetic flux

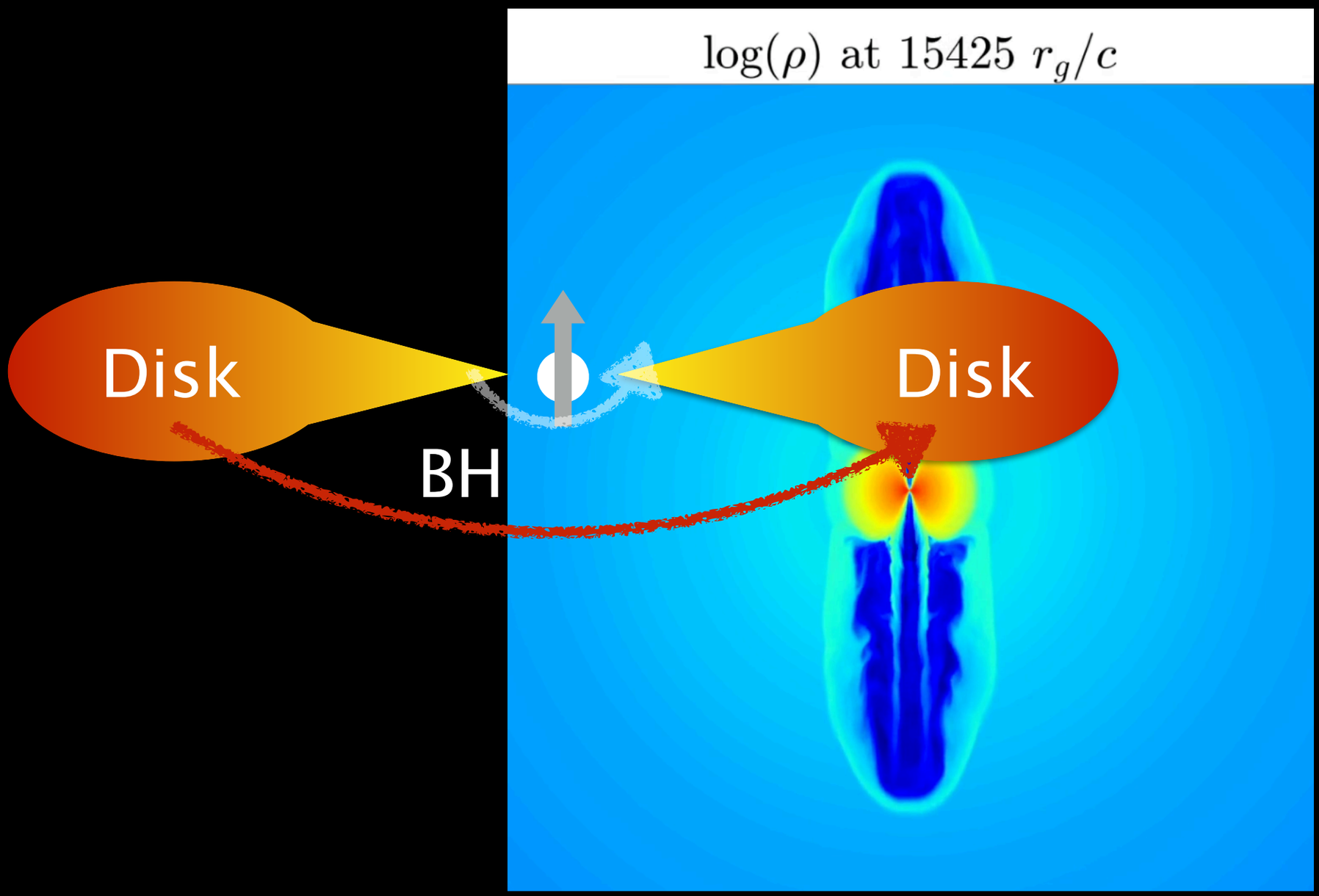
Dynamo turbulence in the disk / interstellar medium? Larger scales?



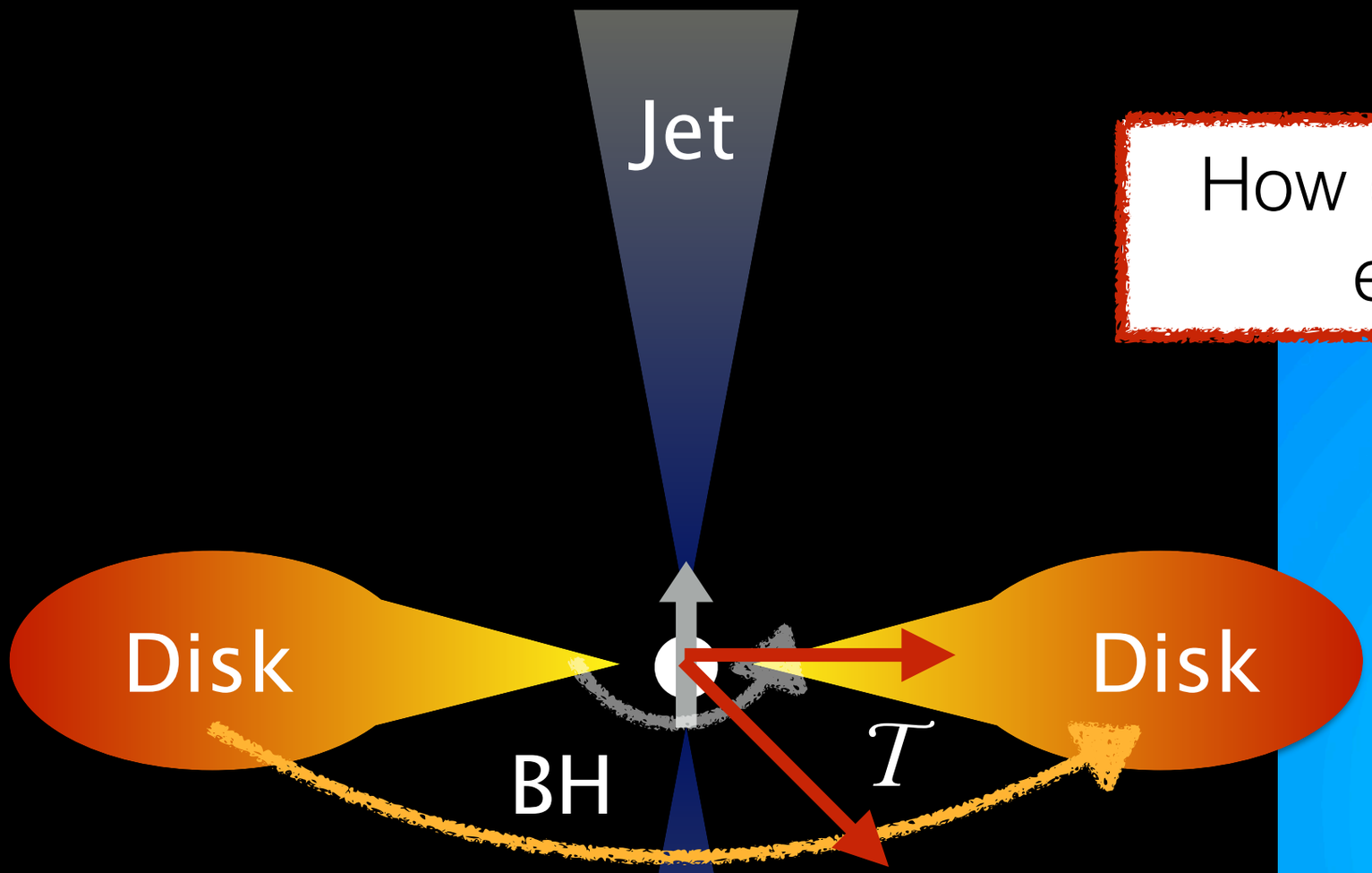
Vertical Magnetic Field advection?

KITP turbulence workshop

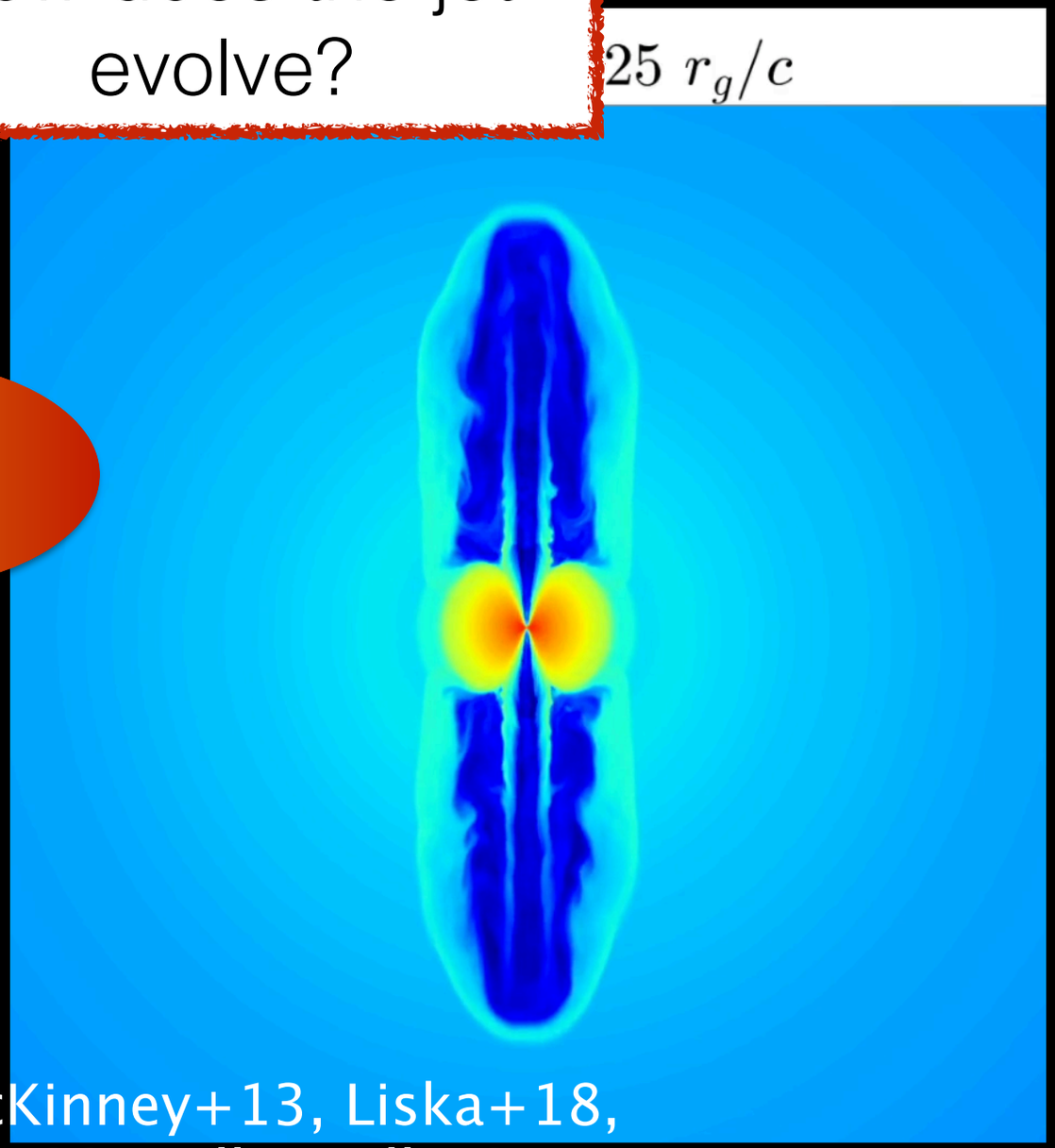
Are MADs universal in starved black holes?



Are MADs universal in starved black holes?



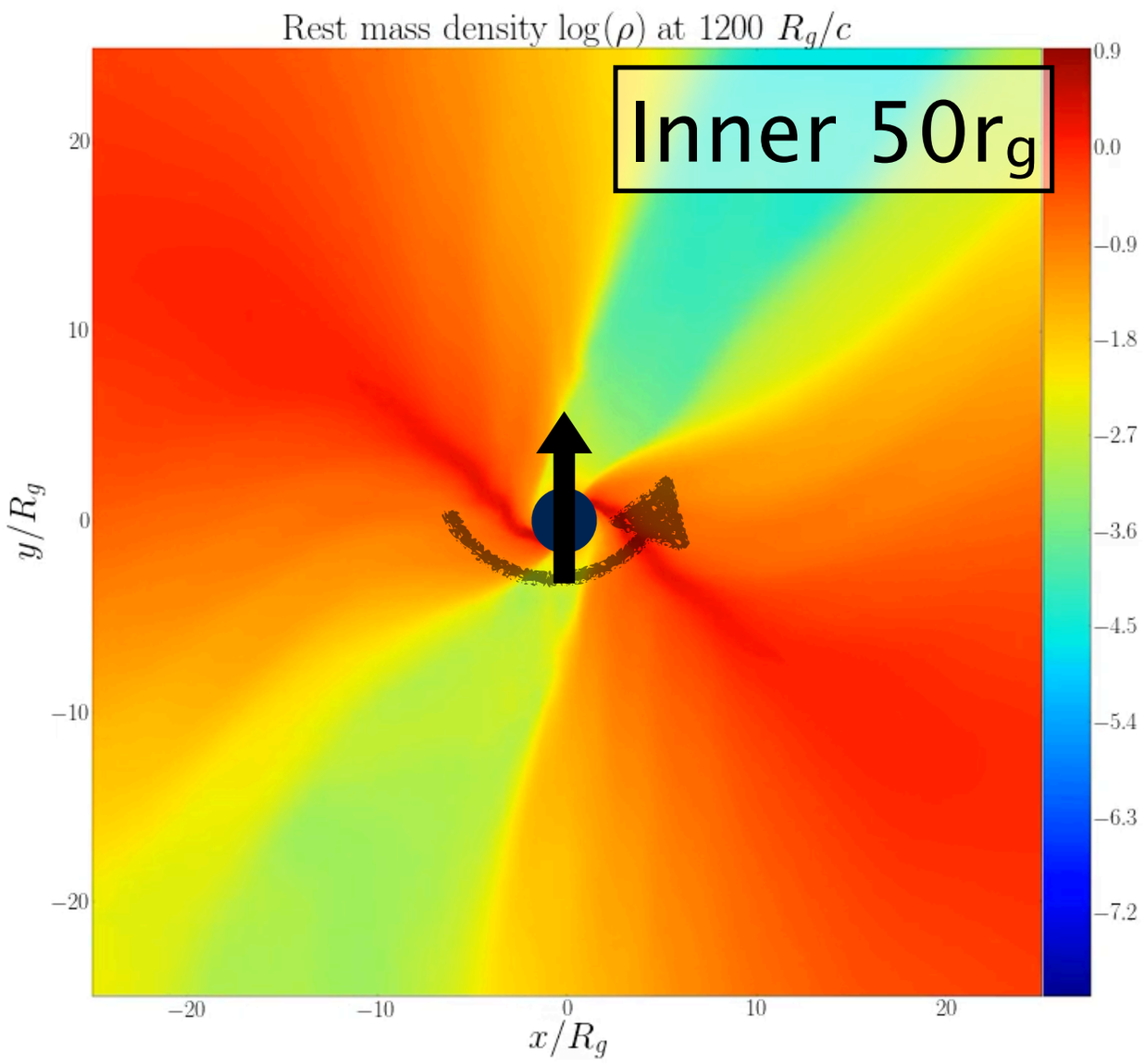
How does the jet evolve?

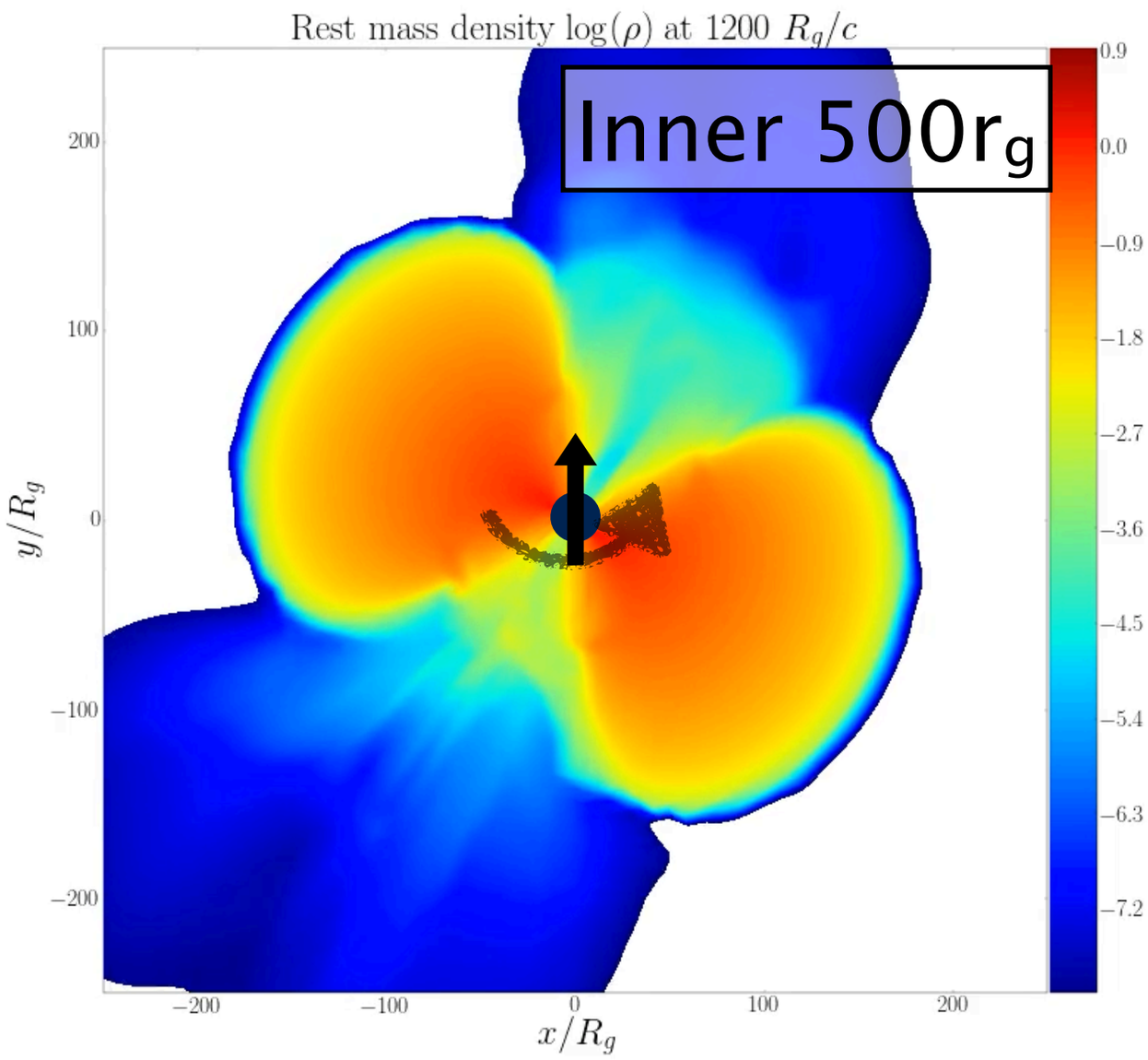


Misaligned disks: Fragile+07, McKinney+13, Liska+18, White+19, **KC**+20, Ressler+21,23, Bollimpalli+24

Misaligned accretion

Liska+18; **KC**+20

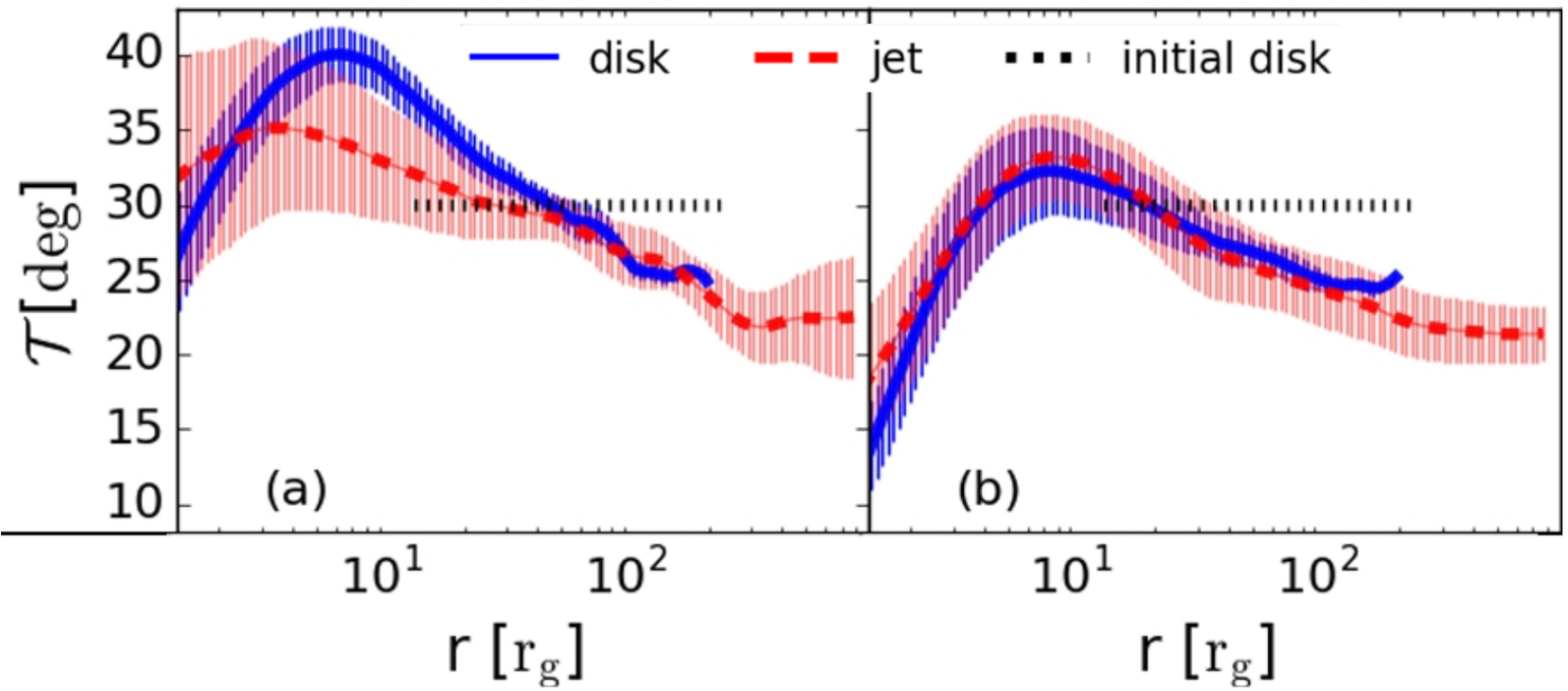




Jet aligns with the disk at large distances!

Disk alignment and magnetic fields

Weak field Strong field



Liska+18, **KC**+20

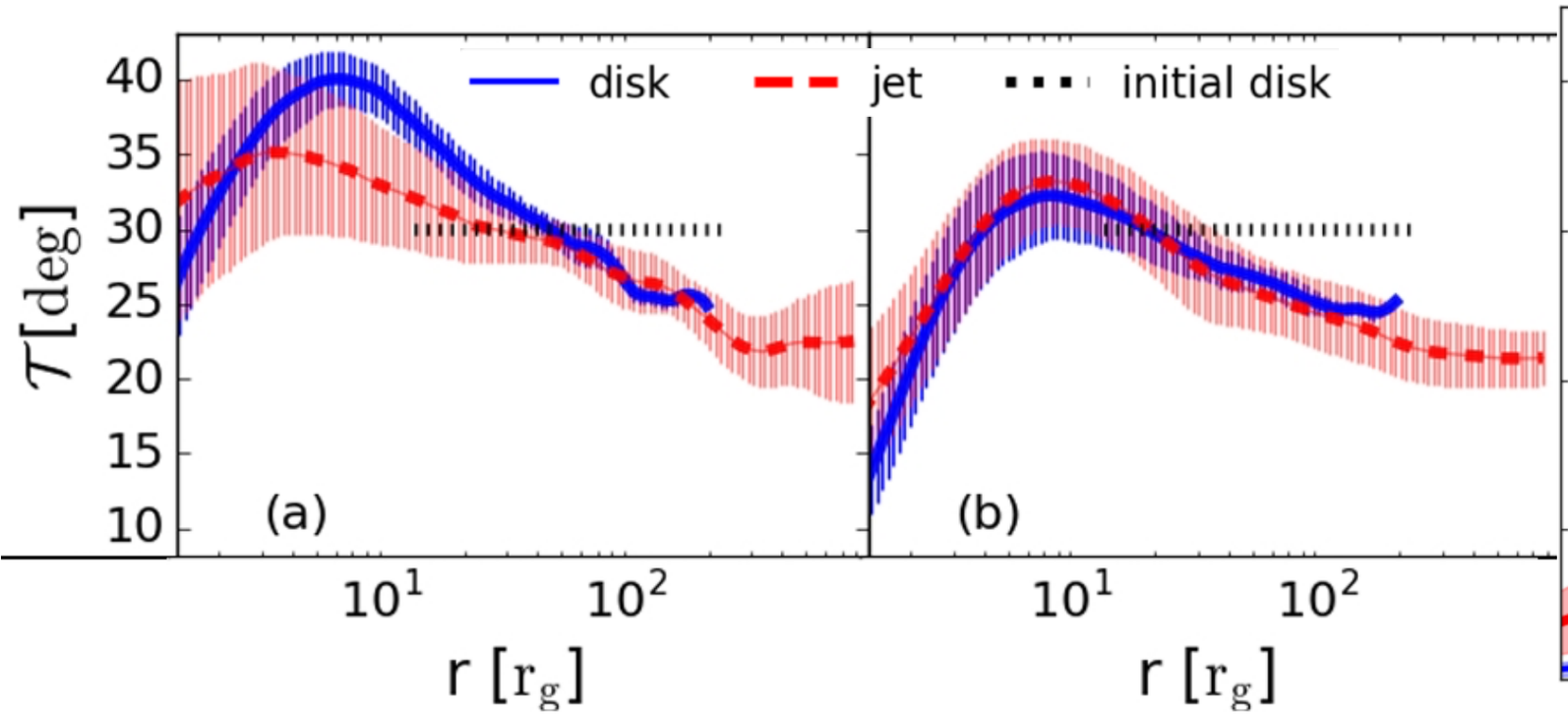
$\mathcal{T}$: deviation from perfect disk-BH alignment

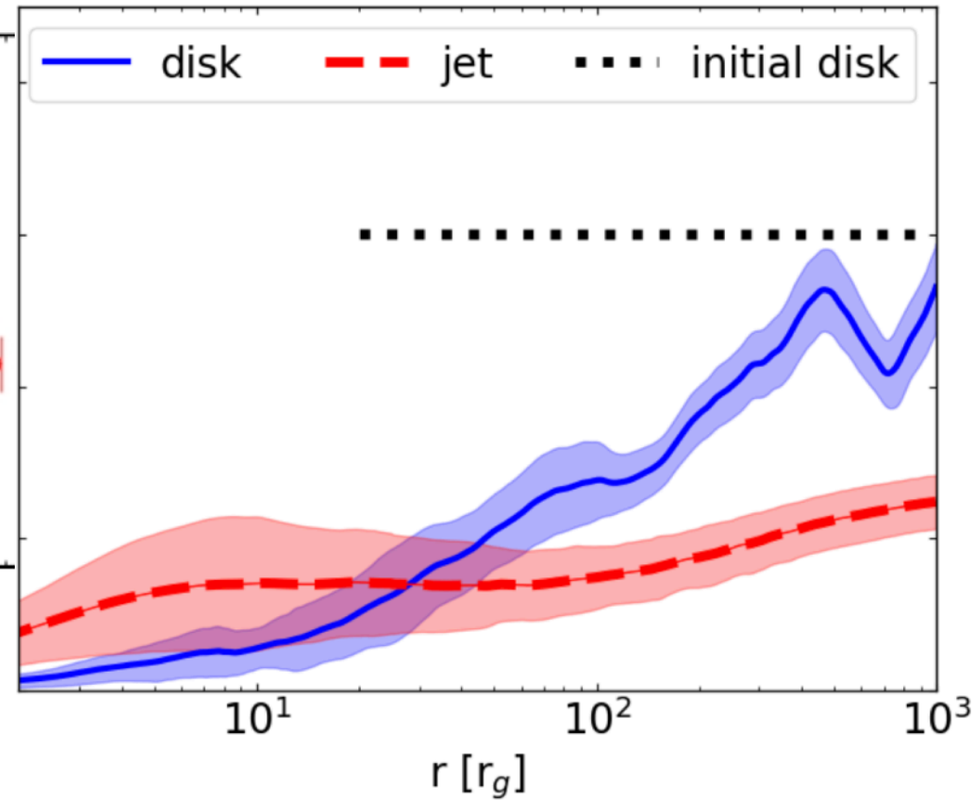
Powerful jets force disk alignment

Weak field

Strong field

MAD field





Liska+18, **KC**+20

$\mathcal{T}$: deviation from perfect disk-BH alignment

McKinney+13,
KC, Liska, Tchekhovskoy+
(arXiv:2311.00432)

Powerful jets force disk alignment

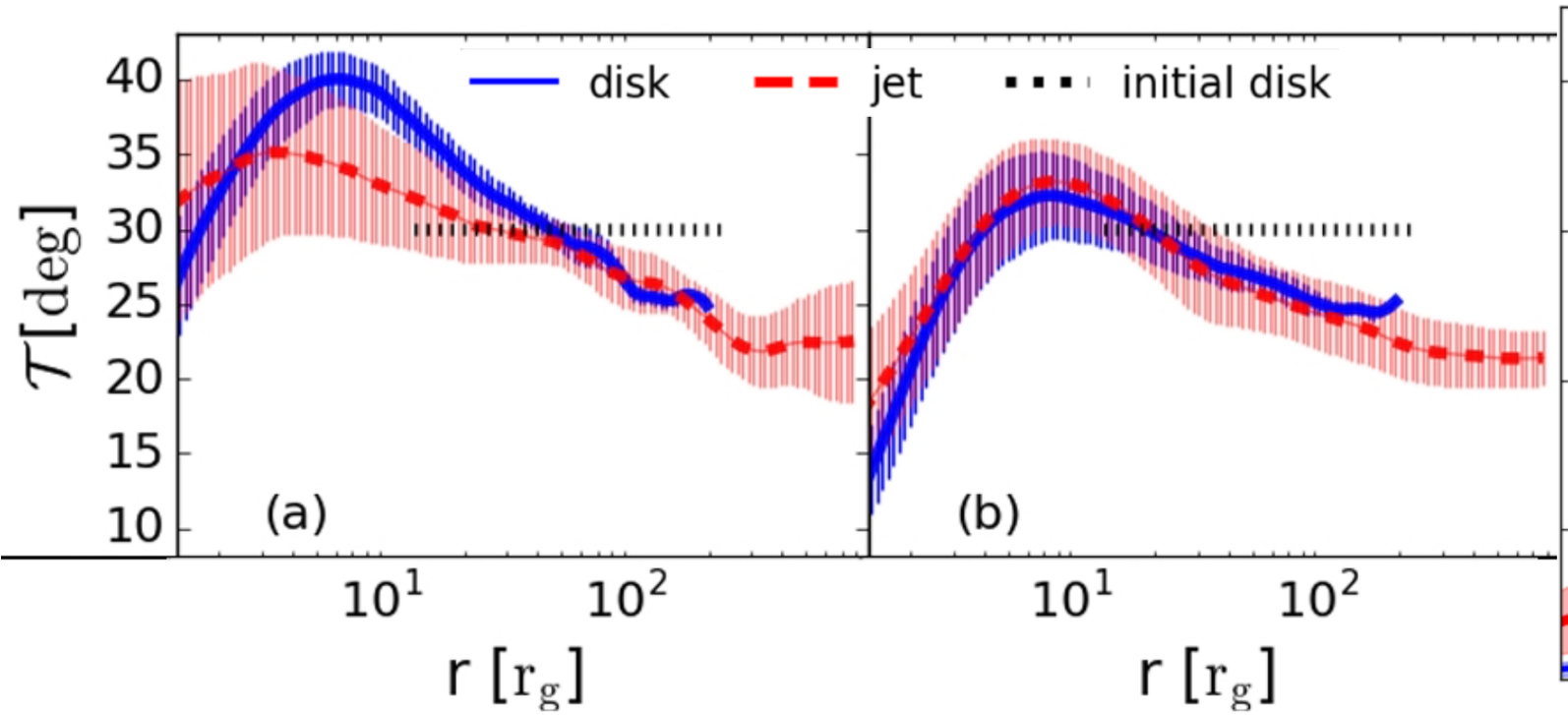
Magneto-spin alignment in MADs

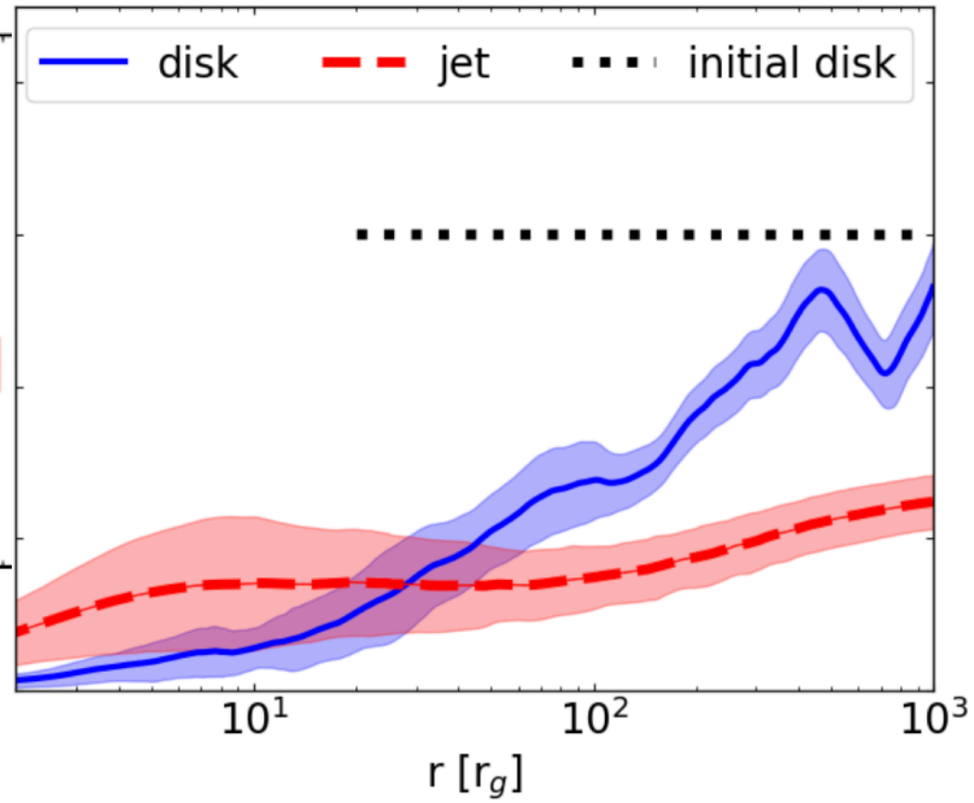
(mechanism different from Bardeen-Petterson alignment)

Weak field

Strong field

MAD field



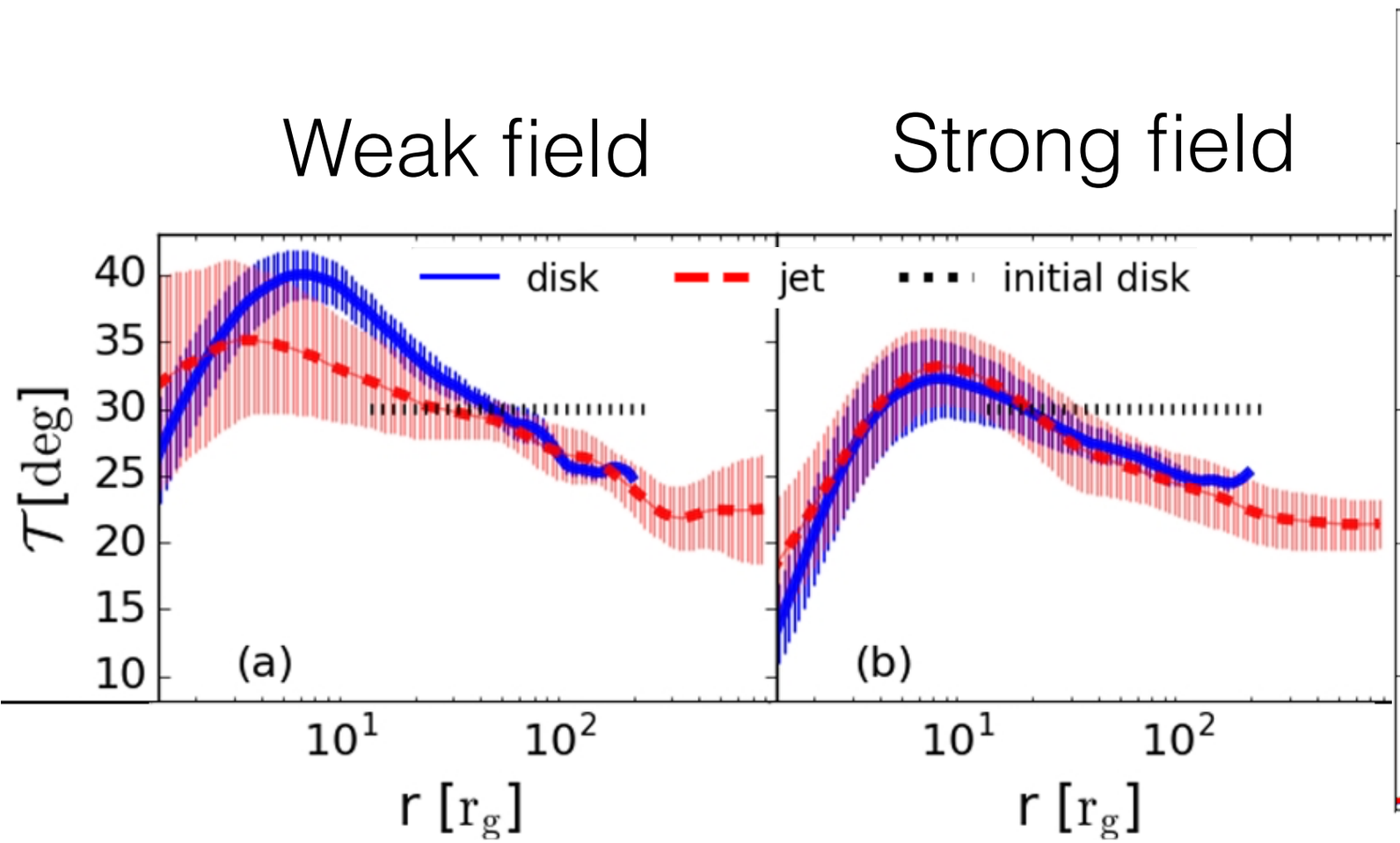


Liska+18, **KC**+20

$\mathcal{T}$: deviation from perfect disk-BH alignment

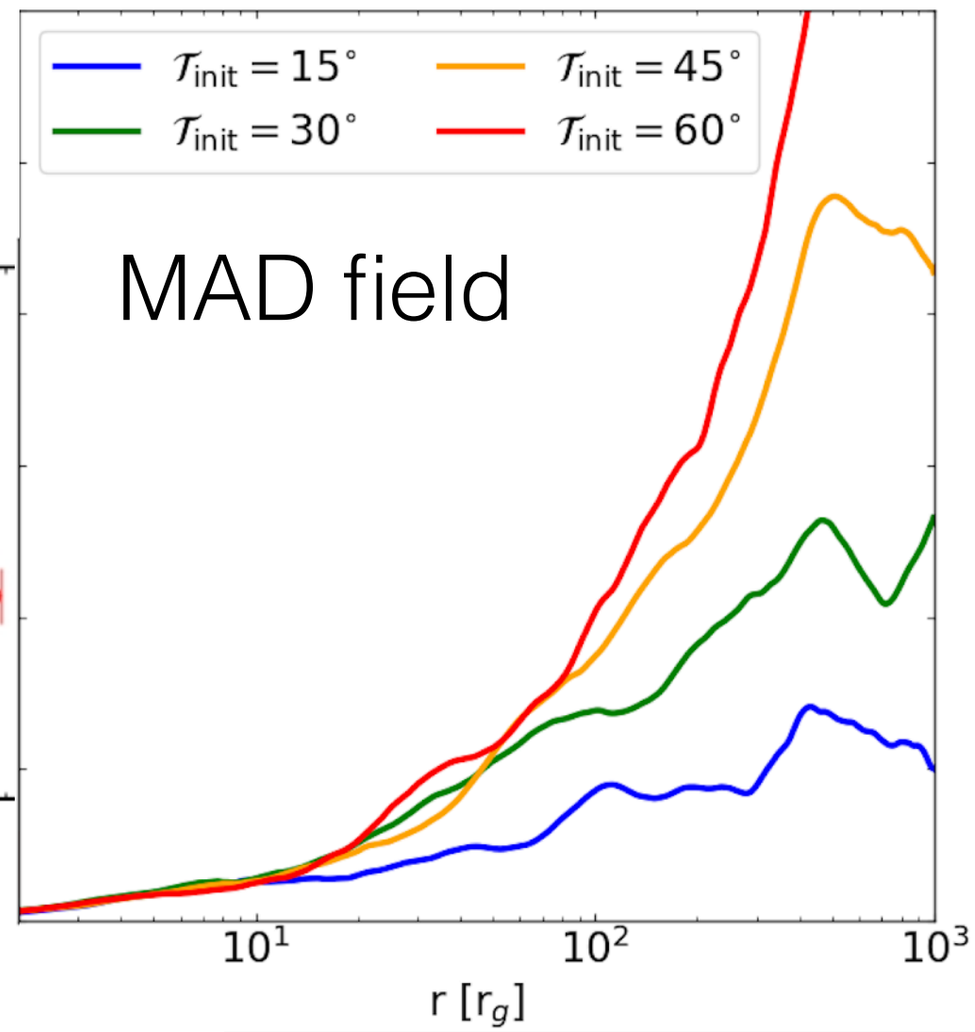
McKinney+13,
KC, Liska, Tchekhovskoy+
(arXiv:2311.00432)

Powerful jets force disk alignment



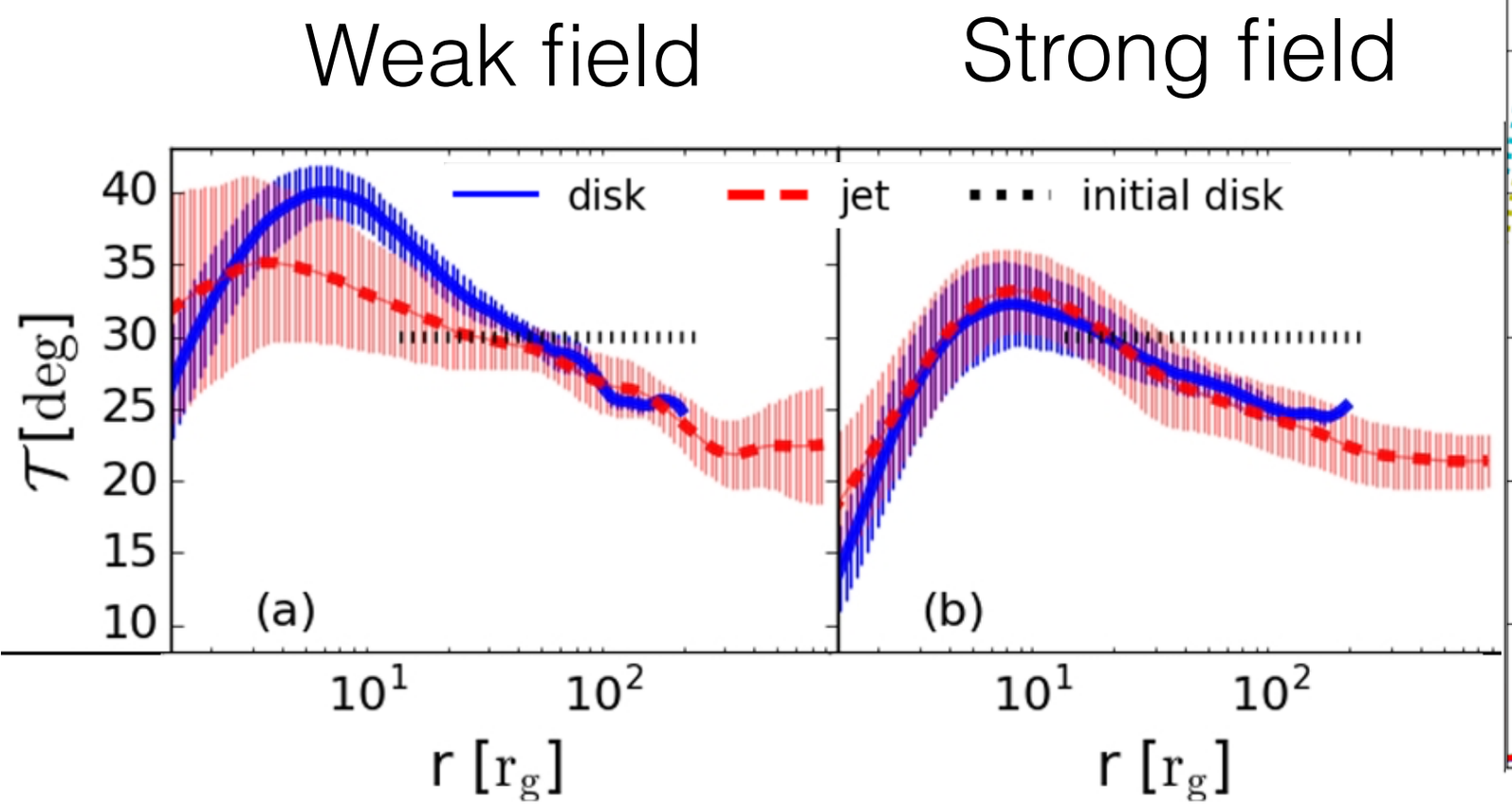
Liska+18, **KC**+20

$\mathcal{T}$: deviation from perfect disk-BH alignment



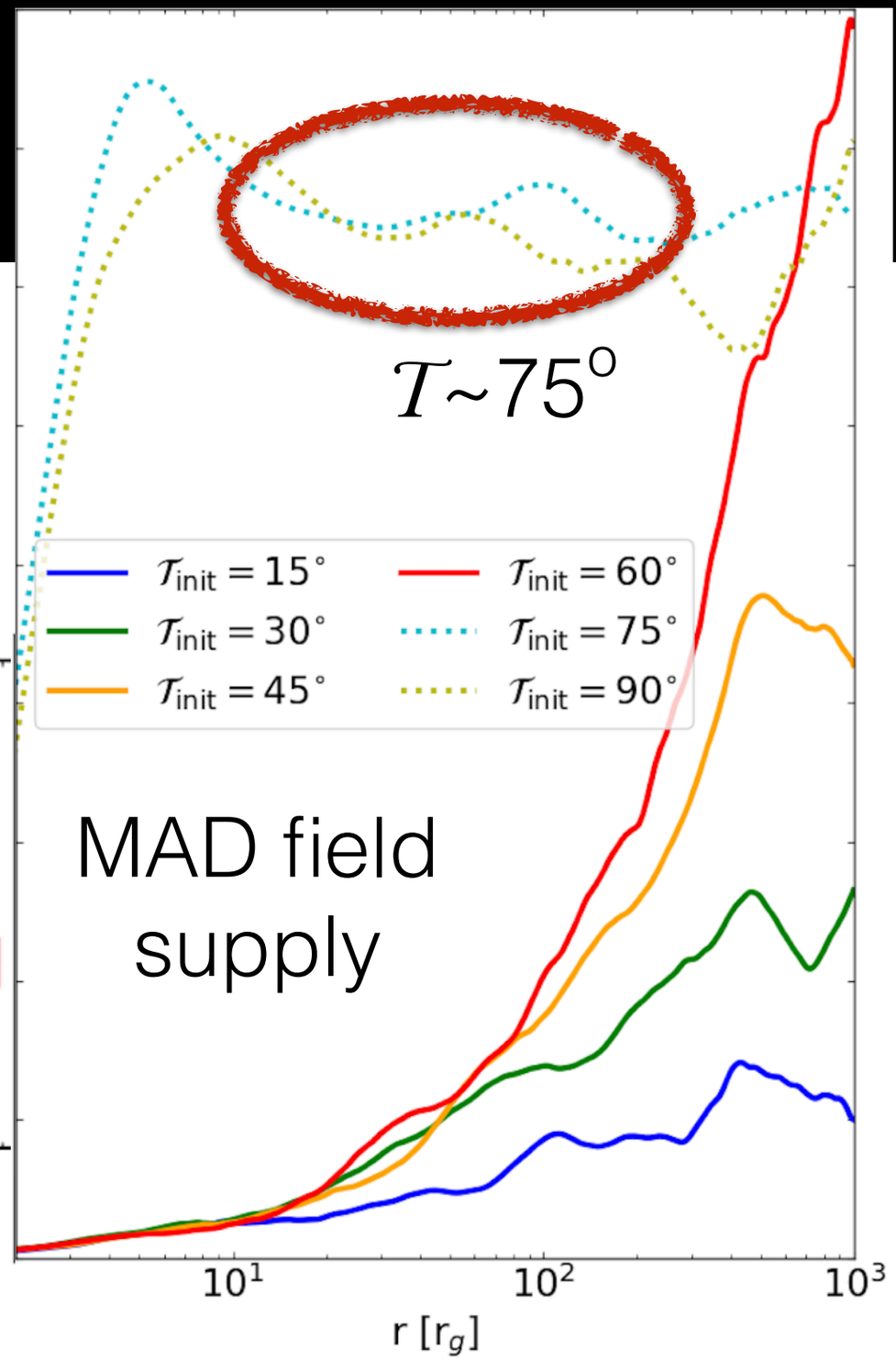
McKinney+13,
KC, Liska, Tchekhovskoy+
(arXiv:2311.00432)

Extreme misalignments



Liska+18, **KC**+20

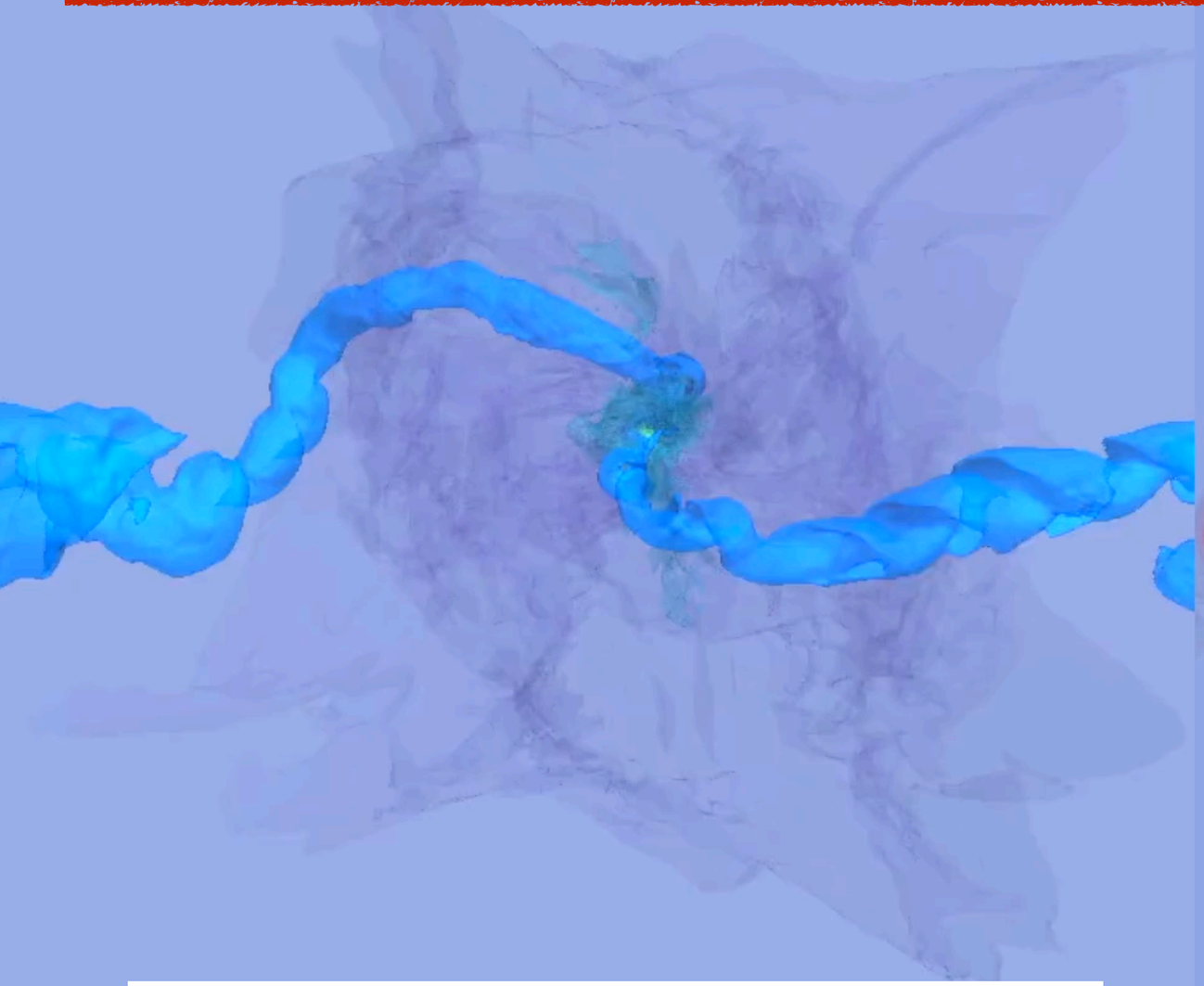
$\mathcal{T}$: deviation from perfect disk-BH alignment



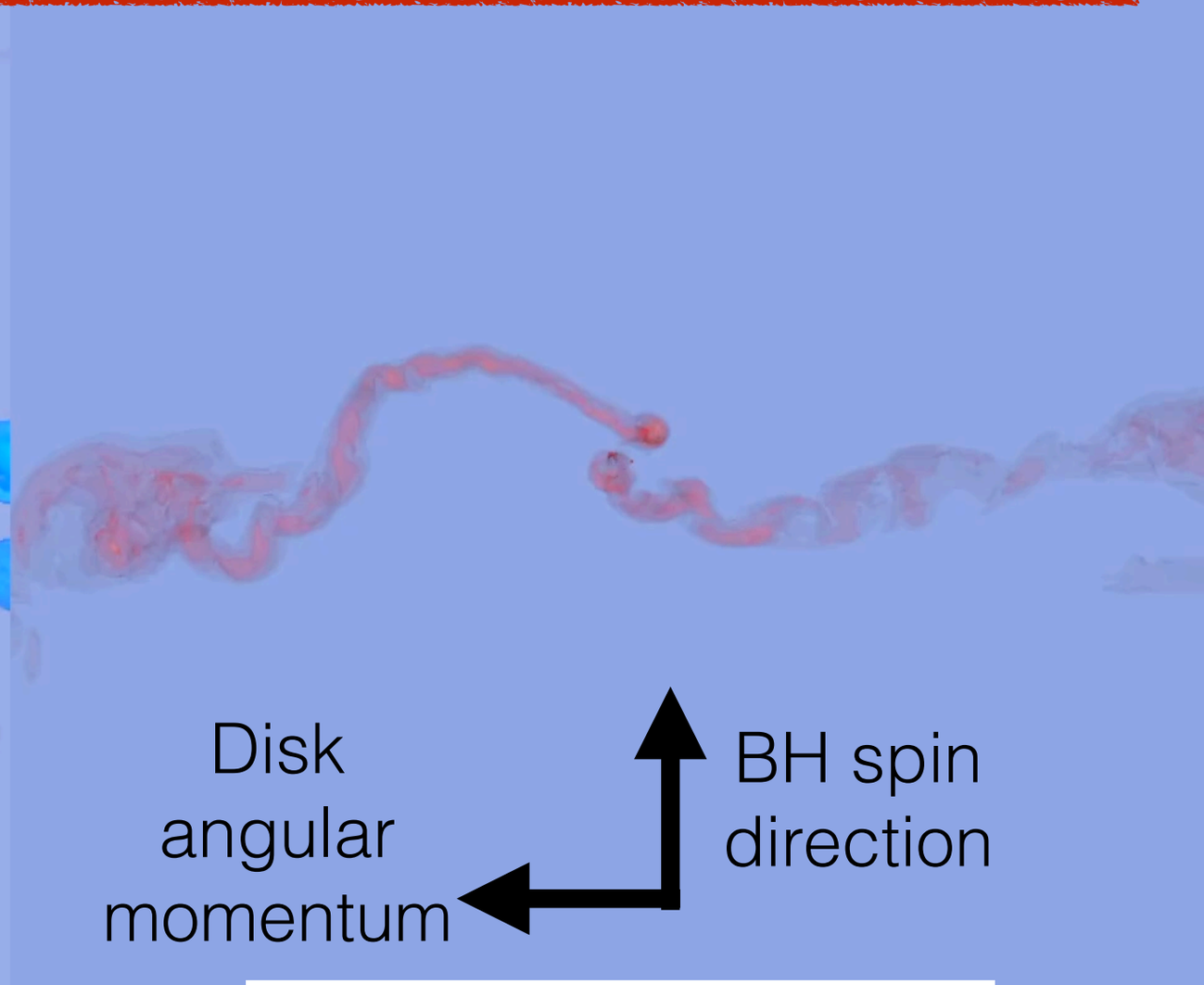
KC, Liska, Tchekhovskoy+ (arXiv:2311.00432)

Evolution of a 90 degrees misaligned disk

Initially the jet launches perpendicular to the black hole spin



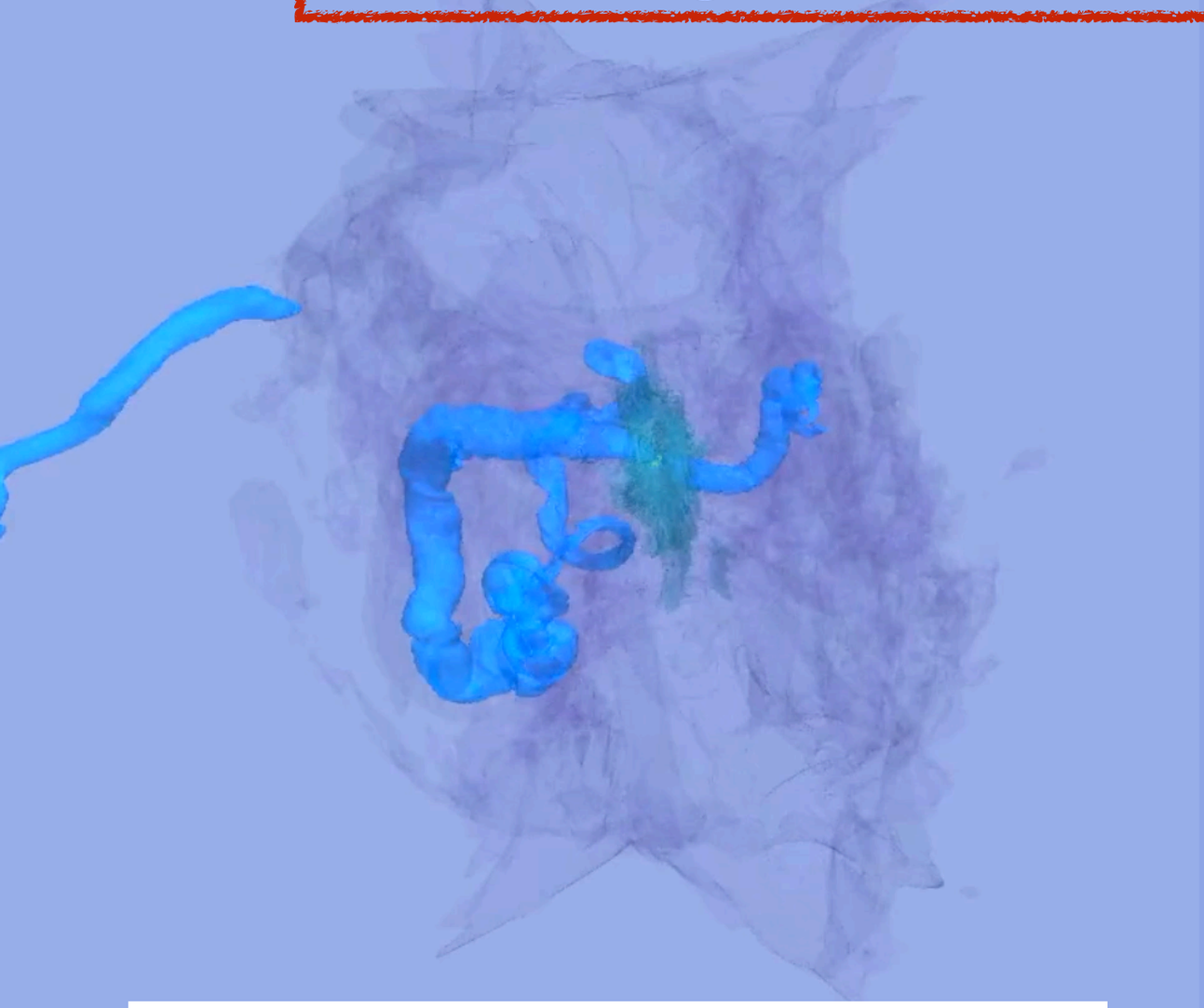
Gas density in ~green-purple (disk)
magnetization in blue (jet)



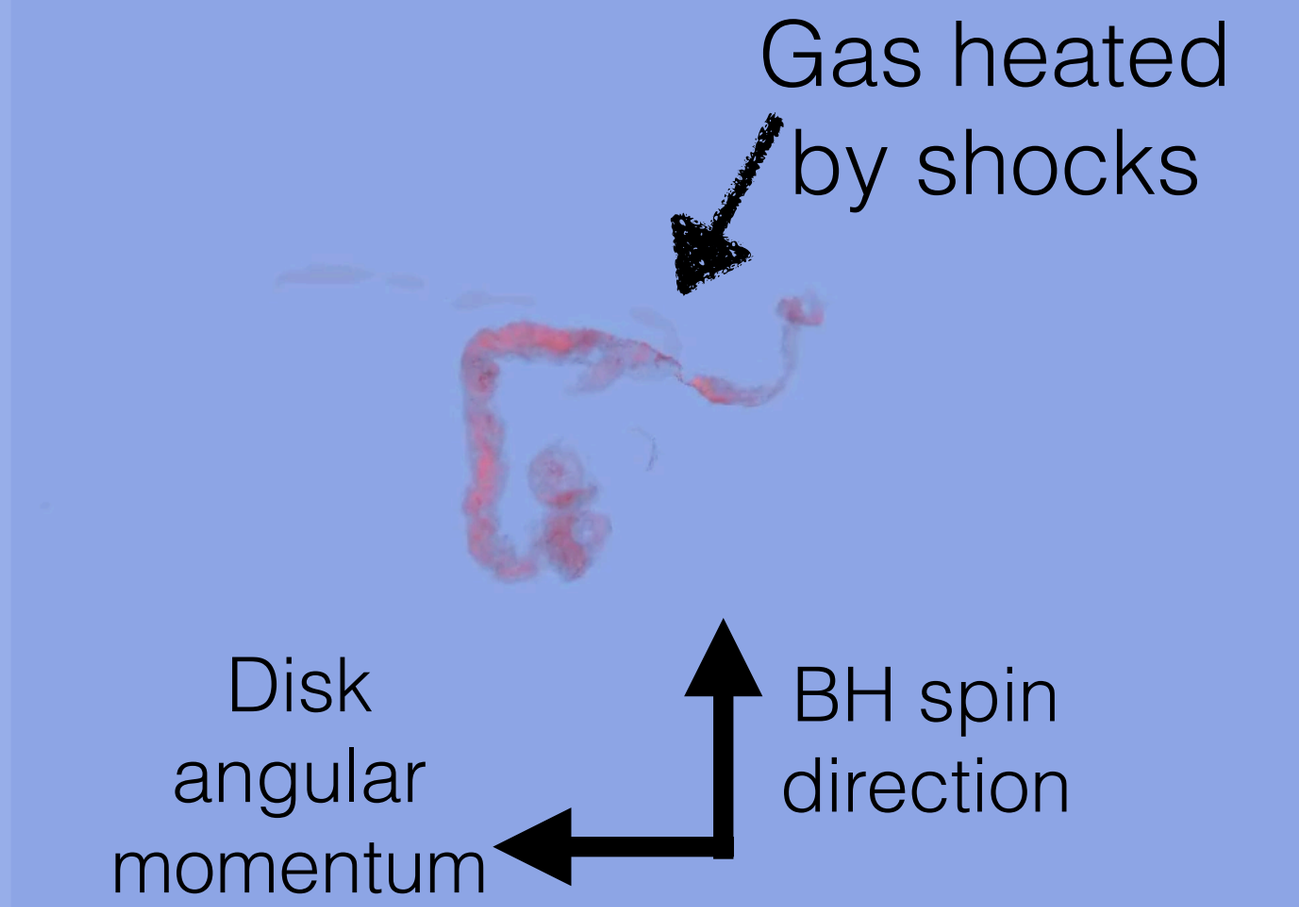
Gas Temperature
(Yellow-orange: relativistic)

Evolution of a 90 degrees misaligned disk

The jet aligns and starts burrowing into the disk



Gas density in ~green-purple (disk)
magnetization in blue (jet)



Gas Temperature
(Yellow-orange: relativistic)

Evolution of a 90 degrees misaligned disk

Jet Quenching & slow restart

Formation of buoyant magnetized filaments & cavities

Disk angular momentum

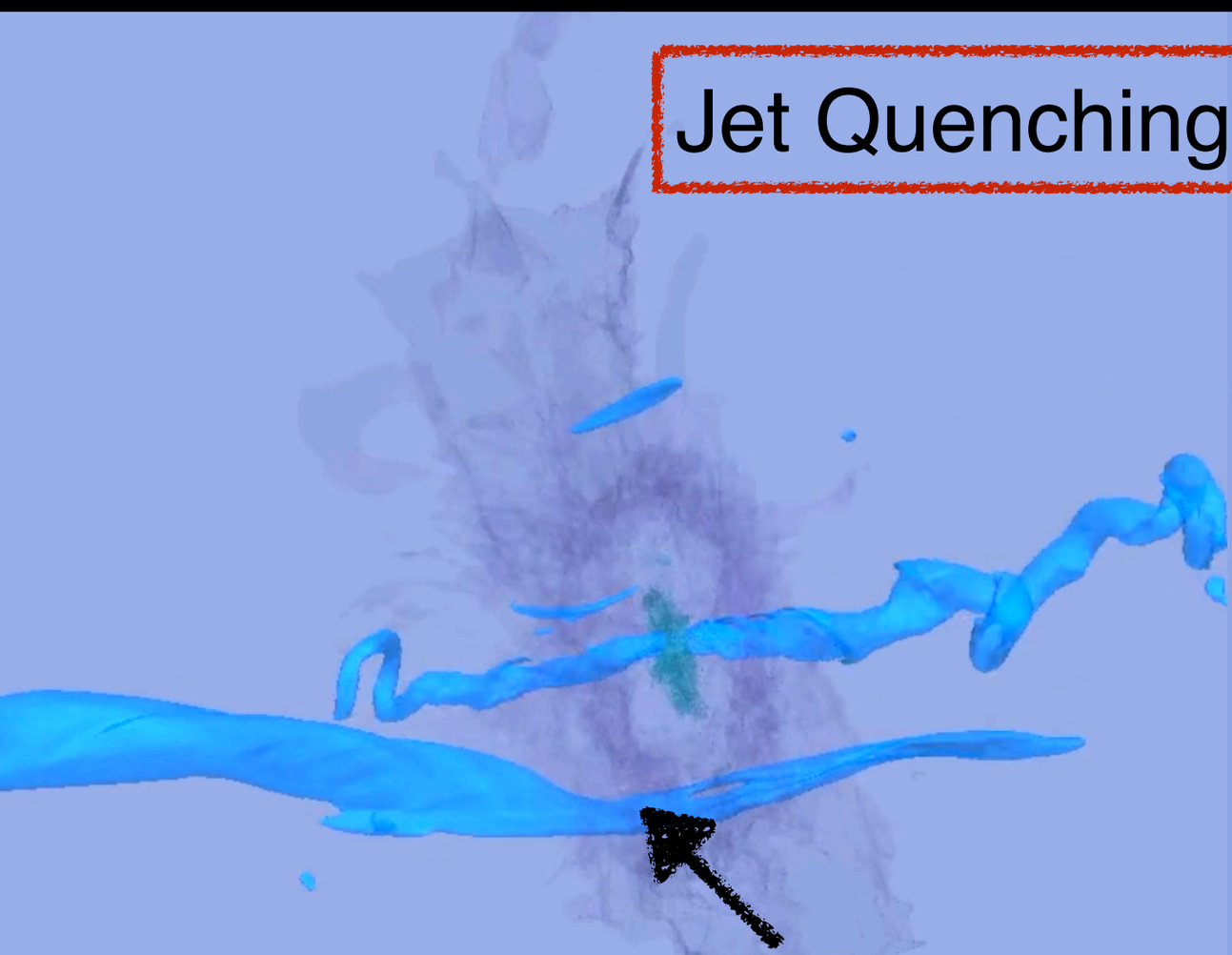
BH spin direction

Gas density in ~green-purple (disk) magnetization in blue (jet)

Gas Temperature (Yellow-orange: relativistic)

Evolution of a 90 degrees misaligned disk

Jet Quenching & slow restart



Formation of buoyant magnetized filaments & cavities

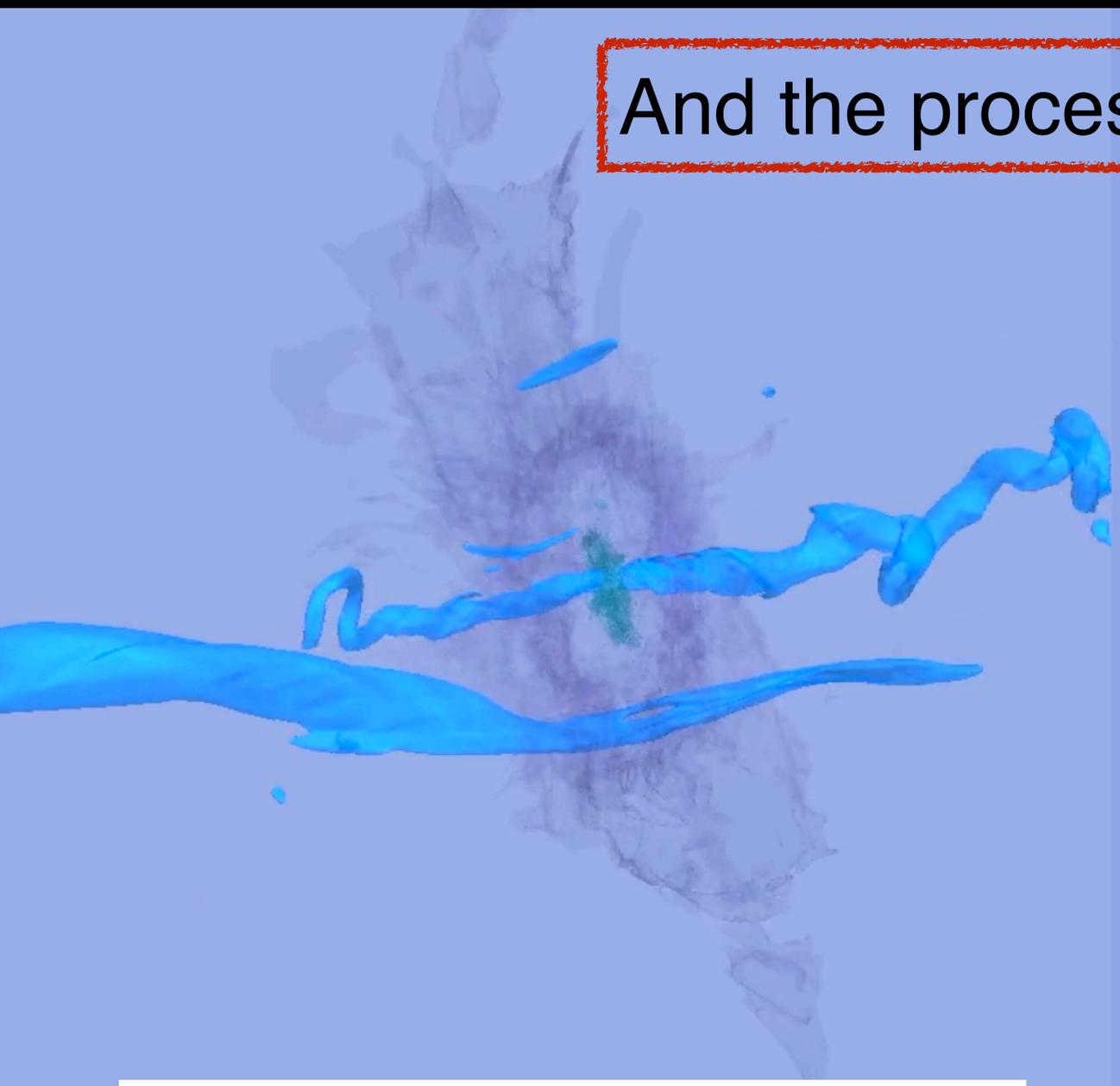
- Filaments seem to follow disk angular momentum:
- Can they stay coherent and explain galaxy-scale filaments?

Gas density in ~green-purple (disk) magnetization in blue (jet)

Gas Temperature (Yellow-orange: relativistic)

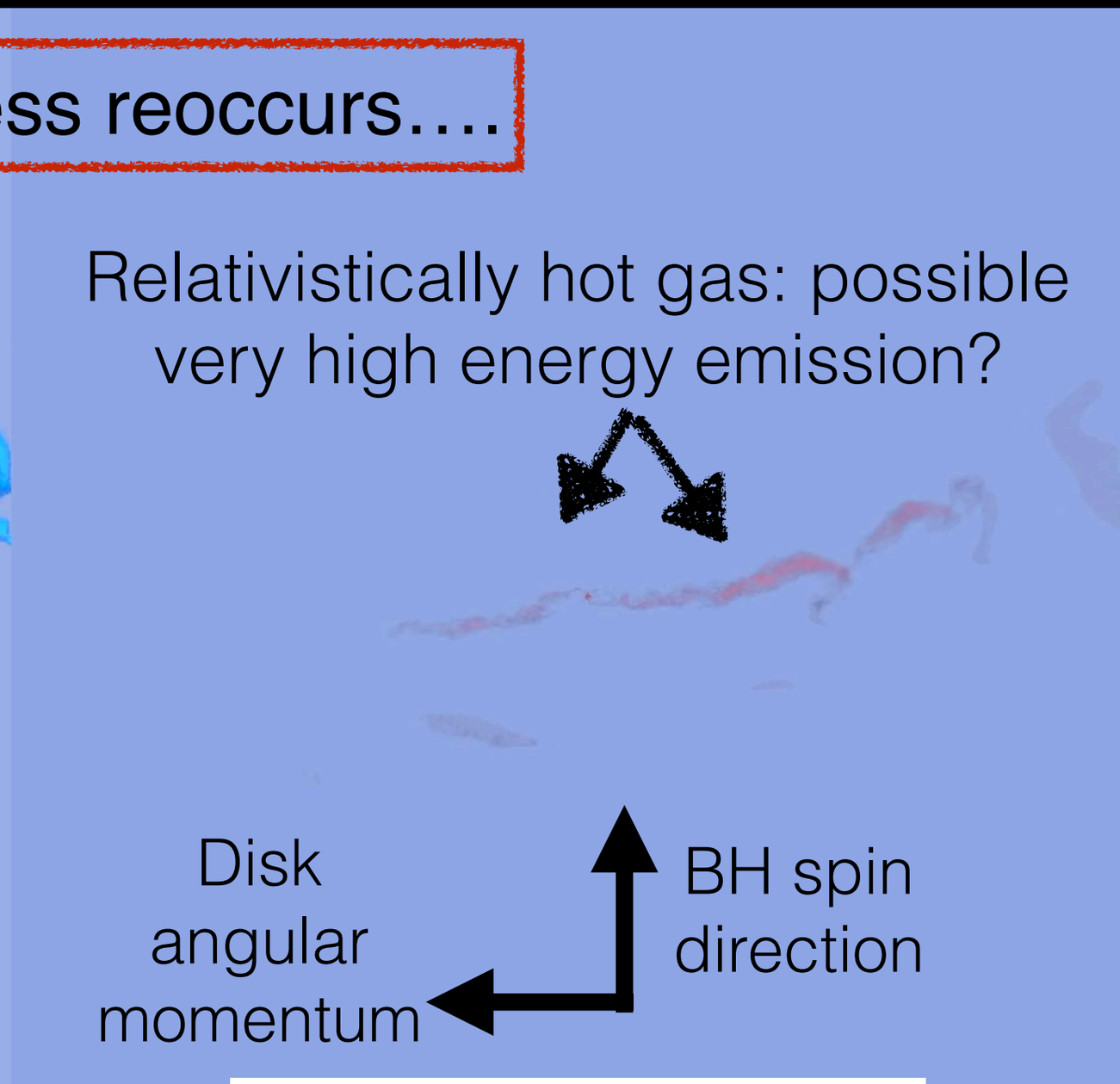
Evolution of a 90 degrees misaligned disk

And the process reoccurs….



Gas density in ~green-purple (disk) magnetization in blue (jet)

Relativistically hot gas: possible very high energy emission?

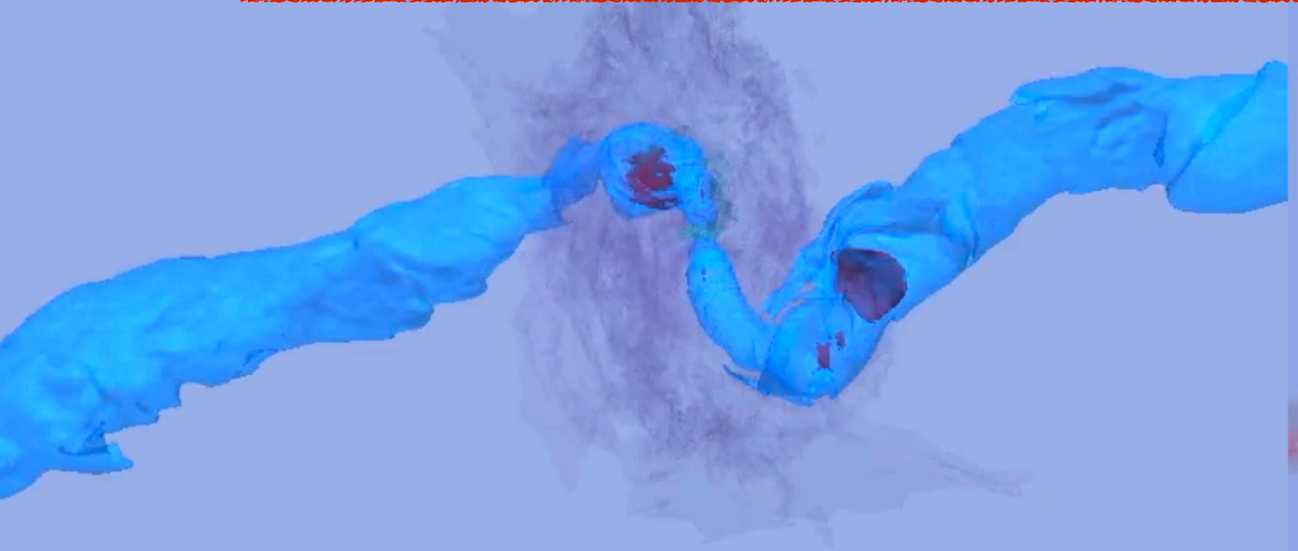


Gas Temperature (Yellow-orange: relativistic)

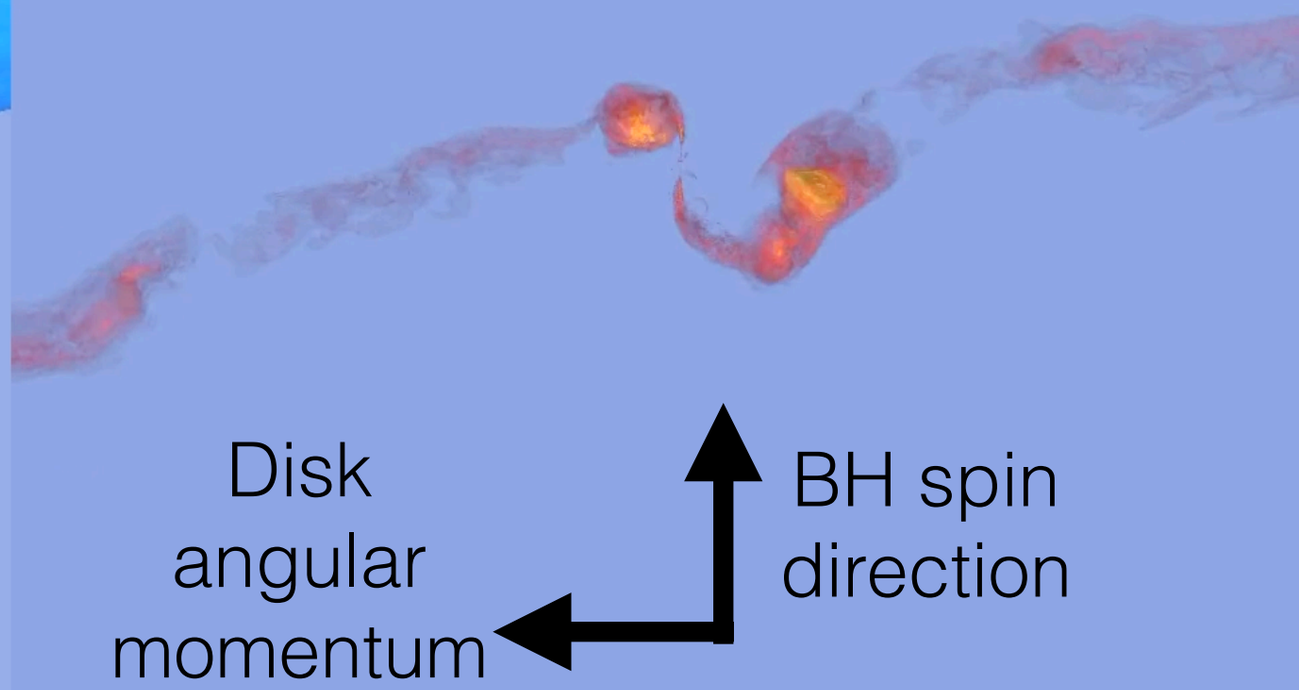
Intermittent jets in extremely misaligned disks

Can this happen in low-luminosity AGN or low state XRBs?

Small-scale Black Hole feedback with period of ~50,000 r_g/c
(~8 yrs for $10^9\,M_\odot$ black hole)

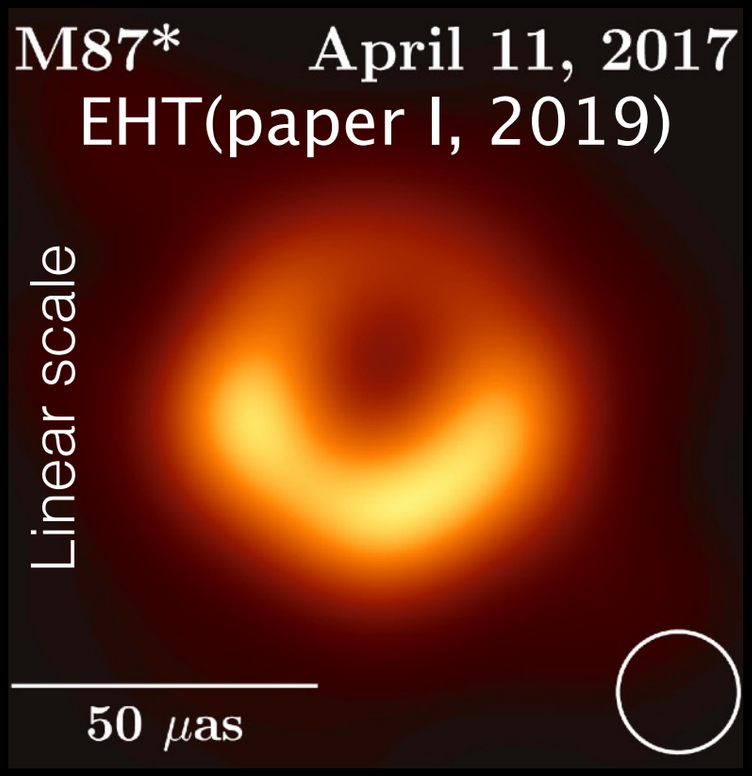


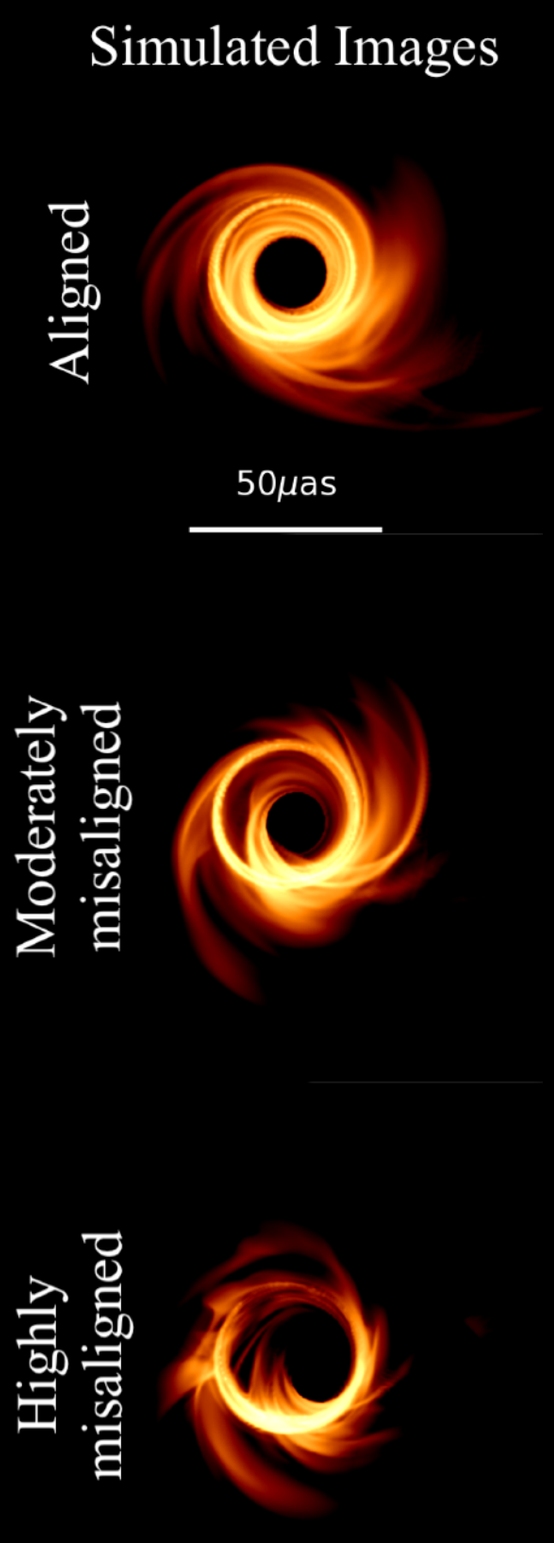
**Gas density in ~green-purple (disk)
magnetization in blue (jet)**



**Gas Temperature
(Yellow-orange: relativistic)**

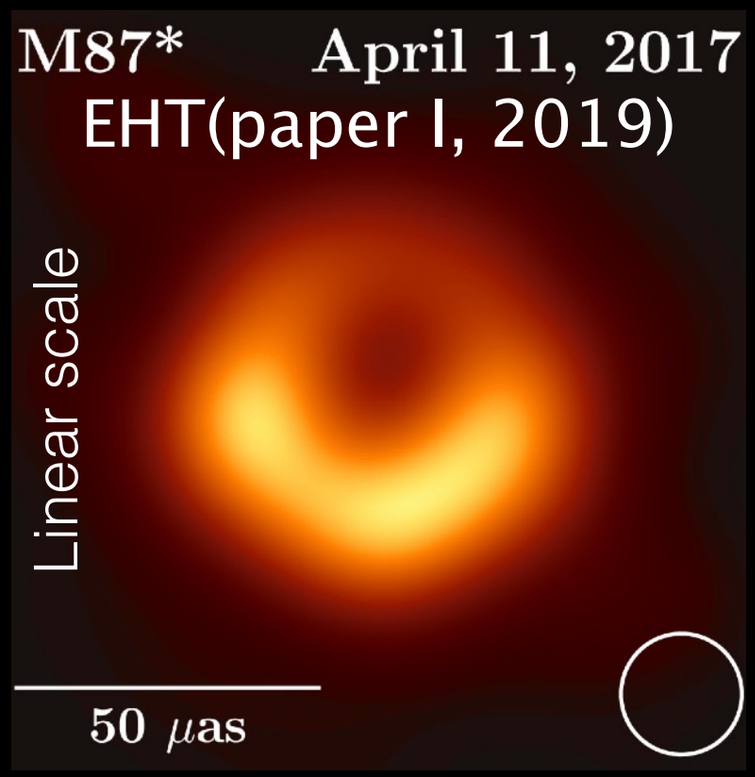
Direct detection of misalignment ?





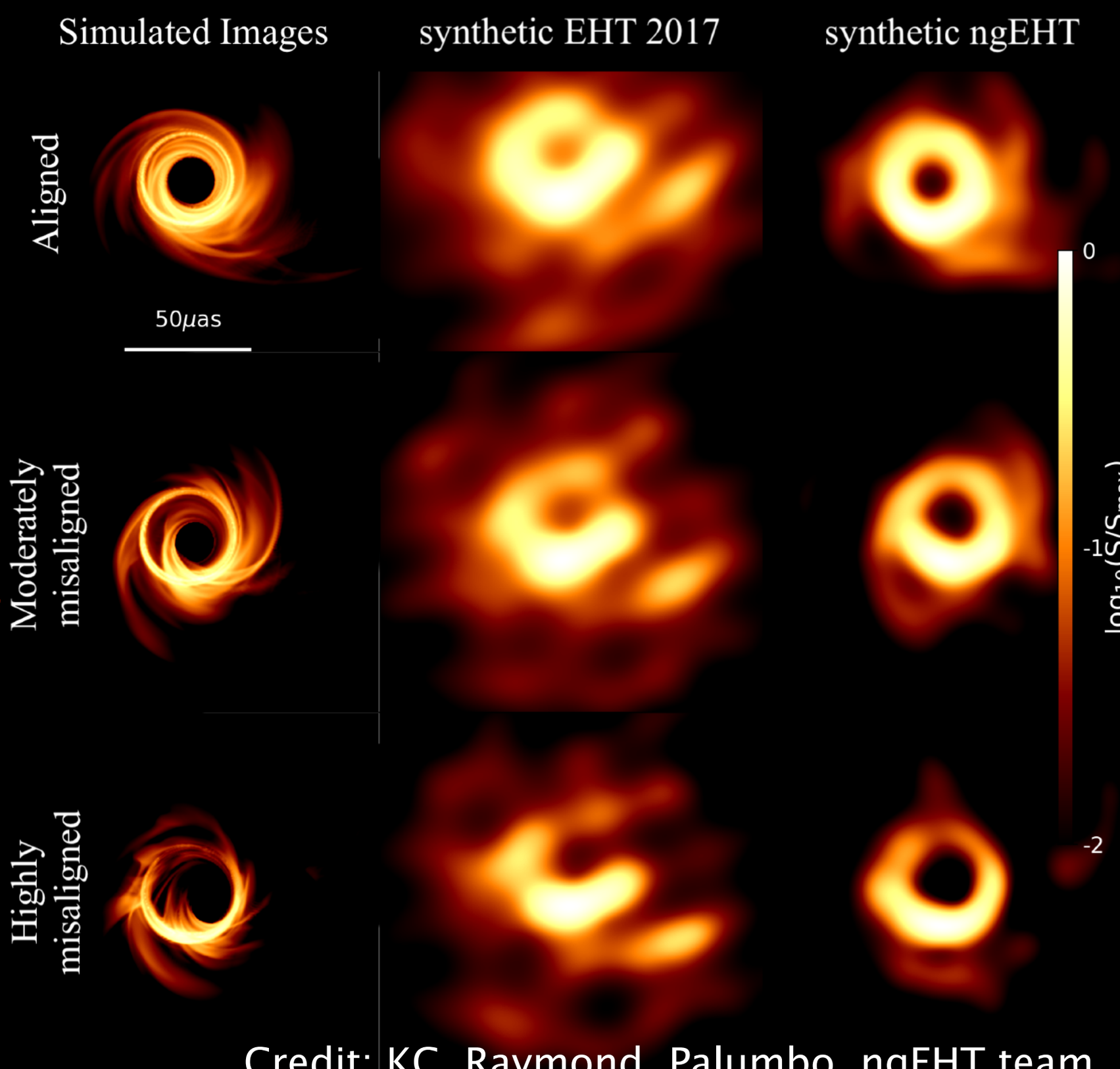
Koushik Chatterjee

Direct detection of misalignment ?



Future versions of EHT could resolve the BH shadow shape & the jet base!

Koushik Chatterjee



Credit: KC, Raymond, Palumbo, ngEHT team

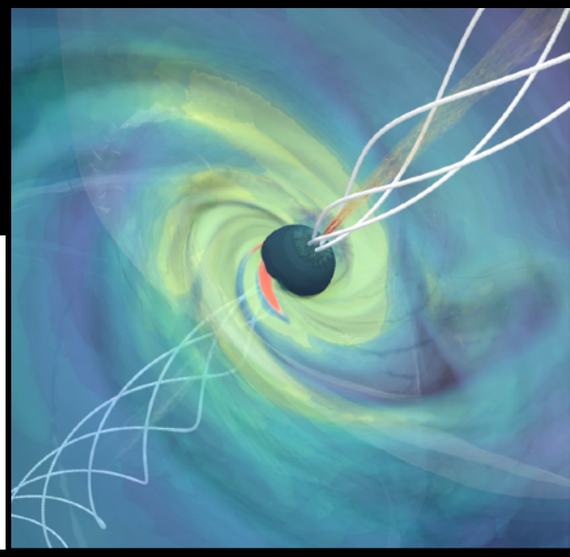
It's a MAD black hole!!!



- Modulations of the magnetic flux determine jet structure and disk alignment: could trigger jet instabilities
  (**KC**+19, Lalakos,..,**KC**+23, Rohoza,..,**KC**+24, **KC**, Liska+; arXiv:2311.00432)

- Extreme disk-BH misalignments produce magnetized filaments, cavities, shocked gas roughly periodically: possible connections to quasi-periodic Xray/gamma-ray emission!
  (**KC**, Liska+; arXiv:2311.00432)

- We are at an exciting stage when BH theory & numerical work and observations are being directly compared: much more to look forward to!!



Koushik Chatterjee